**PRELIMENERY REPORT ON**

# SMARTRESQ+: AN IOT-BASED EMERGENCY RESCUE SYSTEM

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE
FOR THE AWARD OF THE DEGREE

OF

## BACHELOR OF ENGINEERING (COMPUTER ENGINEERING)

**SUBMITTED BY**


| | |
|---|---|
| SHARVARI CHAMBHARE | B400260077 |
| SAI DESHMUKH | B400260086 |
| NEHA SHINGE | B400260193 |

## DEPARTMENT OF COMPUTER ENGINEERING

**PDEA's COLLEGE OF ENGINEERING MANJARI (Bk.),**

**MANJARI, PUNE 412307**


**SAVITRIBAI PHULE PUNE UNIVERSITY**
**2025-26**

# CERTIFICATE

This is to certify that the project report entitles

**"SMARTRESQ+: AN IOT-BASED EMERGENCY RESCUE SYSTEM"**

Submitted by

| | |
|---|---|
| SHARVARI CHAMBHARE | B400260077 |
| SAI DESHMUKH | B400260086 |
| NEHA SHINGE | B400260193 |

is a bonafide student of this institute and the work has been carried out by him/her under the supervision of **Dr. M P. Borawake** and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree of **Bachelor of Engineering** (Computer Engineering).

**Dr. M.P.Borawake**
Guide
Department of Computer Engineering

**Dr. R.V.Patil/Dr.M.P.Borawake**
Head,
Department of Computer Engineering

**Prof. S. V. Shinde**
Project Coordinator

**External Examiner**

**Dr. R. V. Patil**
Principal,
PDEA's College of Engineering, Manjari (Bk.), Pune – 307

Place: Pune
Date:

# ACKNOWLEDGEMENT

It gives us pleasure in presenting the project report on **'SmartResQ+: An IOT-based Emergency Resque System'**. Firstly, we would like to express our indebtedness appreciation to our guide **Dr. M P. Borawake** her constant guidance and advice played very important role in making the execution of the report. She always gave us her suggestions that were crucial in making this report as flawless as possible.

We would like to express our gratitude towards **Dr. R.V. Patil / Dr. M P. Borawake** Head of Computer Engineering Department and Principal P.D.E.A. College of Engineering

**Dr. R.V. Patil** Sir for his kind co-operation and encouragement which helped us during the completion of this report.

Also, we wish to thank to all faculty members for their whole-hearted co-operation for completion of this report. We also thank our laboratory assistants for their valuable help in laboratory.

**NAME OF STUDENTS**

**1. Sharvari Chambhare**

**2. Sai Deshmukh**

**3. Neha Shinge**

# ABSTRACT

Road safety has become one of the most critical concerns in modern transportation. Thousands of lives are lost each year due to delayed medical assistance following road accidents. The SmartResQ – IoT-Based Emergency Rescue System is designed to address this issue by enabling automatic accident detection and instant alert transmission to emergency responders. The primary objective of this project is to minimize the time gap between the occurrence of an accident and the arrival of help, thereby increasing the chances of saving lives.

The system is centered around an ESP32 microcontroller, which integrates various sensors and modules. An accelerometer detects sudden changes in velocity or orientation during collisions, while a flame sensor identifies fire outbreaks caused by severe impacts. Upon detection, the ESP32 initiates an emergency protocol: the GPS module captures the exact accident location, and the GSM module transmits an alert message containing details such as coordinates, time, and accident type to pre-registered contacts, including hospitals, police stations, and the victim's family. Additionally, a buzzer is activated locally to alert nearby individuals for immediate assistance.

Compact, cost-effective, and easy to install, the system provides a practical real-time accident response solution. By integrating sensing, communication, and alert mechanisms into a single automated platform, SmartResQ demonstrates the potential of embedded systems and IoT technologies in enhancing road safety and building a smarter, more responsive emergency management framework.

Keywords: *Emergency alert system; ESP32; GPS-enabled tracking; GSM-based notification system; Internet of Things (IoT)*

# TABLE OF CONTENTS


| Sr. No. | | Title of Chapter | | Page No. |
|---|---|---|---|---|
| **01** | | **Introduction** | | 1-5 |
| | 1.1 | Overview | | 1 |
| | 1.2 | Motivation | | 1 |
| | 1.3 | Problem Definition and Objectives | | 2 |
| | 1.4 | Project Scope & Limitations | | 3-4 |
| | 1.5 | Methodologies of Problem solving | | 4-5 |
| **02** | | **Literature Survey** | | 6-7 |
| **03** | | **Software Requirements Specification** | | 8-16 |
| | 3.1 | Assumptions and Dependencies | | 8 |
| | 3.2 | Functional Requirements | | 8 |
| | 3.3 | External Interface Requirements (If Any) | | 9 |
| | | 3.3.1 | Hardware Interfaces | 9 |
| | | 3.3.2 | Software Interfaces | 10 |
| | | 3.3.3 | Communication Interfaces | 10 |
| | 3.4 | Nonfunctional Requirements | | 11 |
| | | 3.4.1 | Performance Requirements | 11 |
| | | 3.4.2 | Safety Requirements | 12 |
| | | 3.4.3 | Security Requirements | 12 |
| | | 3.4.4 | Software Quality Attributes | 12 |
| | 3.5 | System Requirements | | 13 |
| | | 3.5.1 | Database Requirements | 13 |
| | | 3.5.2 | Software Requirements (Platform Choice) | 14 |
| | | 3.5.3 | Hardware Requirements | 14 |
| | 3.6 | Analysis Models: SDLC Model to be applied | | 15-16 |
| **04** | | **Project Plan** | | 17-25 |
| | 4.1 | Project Estimate | | 17 |

| | | | | |
|---|---|---|---|---|
| | | 4.1.1 | Reconciled Estimates | 17 |
| | | 4.1.2 | Project Resources | 17 |
| | 4.2 | Risk Management | | 18 |
| | | 4.2.1 | Risk Identification | 18 |
| | | 4.2.2 | Risk Analysis | 19 |
| | | 4.2.3 | Overview of Risk Mitigation, Monitoring, Management | 19 |
| | 4.3 | Project Schedule | | 20 |
| | | 4.3.1 | Project Task Set | 20 |
| | | 4.3.2 | Task Network | 21 |
| | | 4.3.3 | Timeline Chart | 22 |
| | 4.4 | Team Organization | | 23 |
| | | 4.4.1 | Team structure | 23 |
| | | 4.4.2 | Management reporting and communication | 24-25 |
| **05** | | **System Design** | | 26-35 |
| | 5.1 | System Architecture | | 26 |
| | 5.2 | Mathematical Model | | 26-28 |
| | 5.3 | Data Flow Diagrams | | 29-30 |
| | 5.4 | Entity Relationship Diagrams | | 30 |
| | 5.5 | UML Diagrams | | 31-35 |
| **06** | | **Project Implementation** | | 36-39 |
| | 6.1 | Overview of Project Modules | | 36 |
| | 6.2 | Tools and Technologies Used | | 37 |
| | 6.3 | Algorithm Details | | 38-39 |
| | | 6.3.1 | Algorithm 1 | 38-39 |
| **07** | | **Software Testing** | | 40-44 |
| | 7.1 | Type of Testing | | 40-43 |
| | 7.2 | Test cases & Test Results | | 44 |
| **08** | | **Results** | | 45-49 |

| | | | |
|---|---|---|---|
| | 8.1 | Outcomes | 45-49 |
| **09** | | **Conclusions** | 50-52 |
| | 9.1 | Conclusions | 50 |
| | 9.2 | Future Work | 50-51 |
| | 9.3 | Applications | 52 |
| | | **Appendix A**: Problem statement feasibility assessment using, satisfiability analysis and NP Hard, NP-Complete or P type using modern algebra and relevant mathematical models. | 53 |
| | | **Appendix B**: Details of paper publication: name of the conference/journal, comments of reviewers, certificate, paper. | 54-68 |
| | | **Appendix** C: Plagiarism Report of project report. | 69 |
| | | **References** | 70-71 |

## LIST OF FIGURES


| FIGURE | ILLUSTRATION | PAGE NO. |
|---|---|---|
| 1.1 | System Overview | 1 |
| 3.6 | Iterative Waterfall Model | 15 |
| 4.2.3 | Timeline Chart | 22-23 |
| 5.1.1 | System Architecture | 26 |
| 5.3.1 | DFD Level – 0 | 29 |
| 5.3.2 | DFD Level – 1 | 29 |
| 5.3.3 | DFD Level – 2 | 30 |
| 5.4 | ER Diagram | 30 |
| 5.5.1 | Activity Diagram Sequence Diagram | 31 |
| 5.5.2 | Class Diagram | 32 |
| 5.5.3 | Object Diagram | 32 |
| 5.5.4 | Usecase Diagram | 33 |
| 5.5.5 | Sequence Diagram | 33 |
| 5.5.6 | State Machine Diagram | 34 |
| 5.5.7 | Component Diagram | 34 |
| 5.5.8 | Deployment Diagram | 35 |
| 5.5.9 | Package Diagram | 35 |
| 8.2.1 | Experimental Setup and Enclosure Design Of Proposed System | 47 |


## LIST OF TABLES


| TABLE | ILLUSTRATION | PAGE NO. |
|---|---|---|
| 4.1.1 | Reconciled Estimates | 17 |
| 7.2 | Test Cases and Test Results | 44 |

# CHAPTER 01: INTRODUCTION

## 1.1 Overview

Road accidents are one of the major causes of injuries and deaths worldwide. In many cases, delayed medical assistance after an accident leads to severe consequences and loss of lives. To address this issue, SmartResQ – IoT-Based Emergency Rescue System is designed as an intelligent safety solution that automatically detects accidents and sends real-time emergency alerts to hospitals, police stations, and family members. The system helps reduce response time and improves the chances of saving accident victims.

The SmartResQ system is developed using an ESP32 microcontroller integrated with an accelerometer sensor, flame sensor, GPS module, GSM module, and buzzer. The sensors detect accidents and fire emergencies, while the GPS and GSM modules provide location tracking and emergency communication. A buzzer also alerts nearby people for immediate assistance. The system is compact, cost-effective, and demonstrates the effective use of IoT and embedded technologies for improving road safety and emergency response systems.

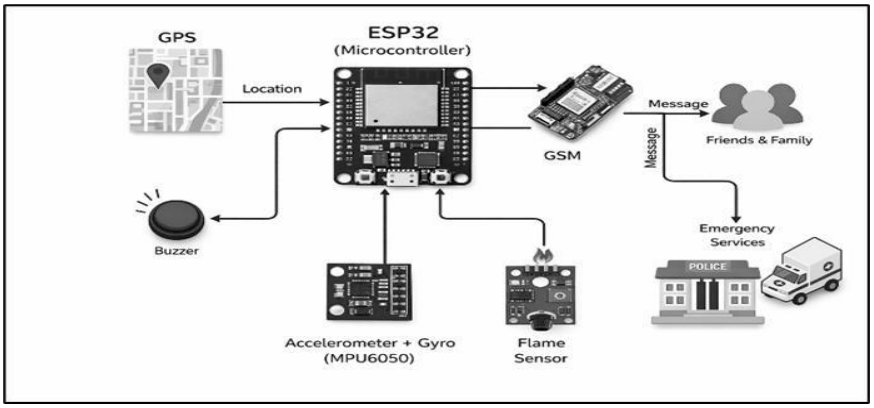



Fig.1.1 System Overview

## 1.2 Motivation

In recent years, road accidents have increased significantly due to rapid urbanization, growing vehicle usage, human errors, and inadequate safety measures. Many accident victims lose their lives because emergency medical assistance does not reach them on time. In several situations, victims are unable to contact emergency services due to

unconsciousness, severe injuries, or remote accident locations. This serious issue became the main motivation behind developing SmartResQ – IoT-Based Emergency Rescue System.

The primary aim of this project is to reduce the time gap between accident occurrence and emergency response by automatically detecting accidents and sending real-time alerts to concerned authorities and family members. By integrating IoT technology, accident detection sensors, GPS tracking, and GSM communication into a single platform, SmartResQ provides a reliable and efficient emergency rescue solution. The project also promotes smarter transportation and improved public safety through automation and real-time communication technologies.

### 1.3 Problem Definition and Objectives

Timely emergency communication plays a very important role in saving lives during road accidents and fire emergencies. However, in many situations, accident victims are unable to seek help due to unconsciousness, severe injuries, or the absence of nearby assistance. The SmartResQ – IoT-Based Emergency Rescue System is developed to address this issue by automatically detecting accidents and fire emergencies using sensors and embedded technologies. The system instantly sends emergency alerts through GSM communication along with real-time GPS location details to hospitals, police stations, and family members. This helps reduce emergency response time and improves the chances of providing quick medical assistance to accident victims.

**Objectives:**

- Develop an Automated Detection System: Build an IoT-based system capable of automatically detecting road accidents and fire emergencies in real time.

- Detect Emergencies Accurately: Identify vehicle collisions and fire accidents using accelerometer and flame sensors with reliable performance.

- Provide Real-Time Location Tracking: Use a GPS module to capture and share the exact accident location coordinates.

• Enable Instant Emergency Alerting: Send automatic emergency alerts and notifications to hospitals, police stations, and family members using GSM communication.

• Improve Emergency Response Time: Reduce the delay between accident occurrence and rescue operations to increase the chances of saving lives.

• Enhance Road Safety Systems: Integrate IoT and embedded technologies to create a compact, reliable, and cost-effective vehicle safety solution.

### 1.4 Project Scope & Limitations

#### 1.4.1 Scope:

• Accident Detection System: The project focuses on detecting road accidents and fire emergencies using sensors and IoT technology.

• Real-Time Monitoring: The system continuously monitors vehicle conditions to identify emergency situations automatically.

• GPS-Based Location Tracking: The project uses a GPS module to provide accurate real-time accident location details.

• Emergency Alert Communication: GSM technology is used to send instant emergency alerts to hospitals, police stations, and family members.

• Vehicle Safety Application: The system can be implemented in cars, buses, ambulances, and other vehicles to improve passenger safety.

• Smart Safety Solution: The project demonstrates the use of IoT and embedded systems for developing intelligent emergency rescue and road safety systems.

### 1.4.2 Limitations:

• GPS Signal Issues: GPS accuracy may decrease in tunnels, underground areas, or places with weak satellite signals.

• Network Dependency: GSM communication depends on mobile network availability and may face delays in poor network regions.

• False Alerts: Sudden jerks or vibrations may sometimes trigger incorrect accident detection alerts.

• Power Requirement: The system requires continuous power supply for proper operation.

• Limited Implementation: The current project is mainly designed as a prototype and may require additional features for large-scale commercial deployment.

## 1.5 Methodologies of Problem Solving

### 1. Requirement Gathering and Analysis

**Activities:**

• Study the problem of delayed emergency response during road accidents and fire emergencies.
• Analyze the need for automatic accident detection and emergency alert systems.
• Define system requirements such as accident detection, GPS tracking, and emergency communication.

### 2. System Design

**Activities:**

• Design the overall architecture and workflow of the SmartResQ system.
• Select components such as ESP32, accelerometer sensor, flame sensor, GPS module, GSM module, and buzzer.
• Plan communication between sensors, controller, and alert modules.

### 3. Hardware and Software Implementation

**Activities:**

• Integrate sensors and communication modules with the ESP32 controller.
• Develop embedded C/C++ code using Arduino IDE for system operation.
• Configure GPS and GSM modules for location tracking and emergency alert transmission.

### 4. Working Procedure / System Workflow

**Activities:**

• Monitor vehicle conditions continuously using sensors.
• Detect accidents and fire emergencies automatically.
• Retrieve accident location through GPS and send emergency alerts using GSM communication.
• Activate the buzzer for nearby emergency assistance.

### 5. Testing and Validation

**Activities:**

• Test accident and fire detection under different conditions.
• Verify GPS tracking and GSM message transmission accuracy.
• Perform overall system testing to ensure reliability and proper emergency response.

# CHAPTER 02: LITERATURE SURVEY

| Sr. No. | Year & Journal | Paper & Author | Methodology | Advantages | Disadvantages |
|---|---|---|---|---|---|
| 1 | 2024, International Journal of Advanced Research in Computer Science | IoT Based Smart Accident Detection and Early Warning System – Babu, C. V. S., & Vinola, M. P. | IoT sensors, GPS, GSM alerts | Real-time alerts, quick response | Network dependency |
| 2 | 2024, IEEE Internet of Things Journal | IoT-Based Smart Emergency Response System (SERS) for Monitoring Vehicle, Home, and Health Status – Mohsin, A. S. M., & Muyeed, M. A. | IoT monitoring, emergency alerts | Multi-purpose safety, automation | High cost, complexity |
| 3 | 2023, International Journal of Intelligent Systems and Applications | AI-Enabled Accident Detection and Alert System Using IoT and Deep Learning – Pathik, N., Sharma, R., & Patel, V. | AI, Deep Learning, IoT | High accuracy, smart detection | Large dataset needed |
| 4 | 2022, International Journal of Computer Applications | Accident Detection and Alerting Systems: A Study – Baballe, M. A. | Comparative study, survey | Research insights, gap analysis | No practical implementation |
| 5 | 2021, Proceedings of the International Conference on | IoT-Based Car Accident Detection and Notification Algorithm – Sharma, S. | IoT algorithm, auto alerts | Fast notification, low human effort | False triggering |

| | | | | | |
|---|---|---|---|---|---|
| | IoT and Smart Systems | | | | |
| 6 | 2018, International Journal of Advanced Computer Science and Applications | Road Accident Detection and Automatic Alert System Using IoT – Amin, M. R., & Jalil, M. A. | GPS, GSM, IoT modules | Automatic alerts, location tracking | GPS/network issues |
| 7 | 2016, International Journal of Computer Applications | Vehicle Accident Detection and Reporting System Using GPS and GSM Module – Chaurasia, P., & Verma, S. | GPS and GSM modules | Simple, cost-effective | Limited smart features |
| 8 | 2021, Journal of Computer Science | IoT Based Message Alert System for Emergency Situations – Alsaad, S. N., & Hussien, N. M. | IoT alert messaging | Rapid communication | Network dependency |
| 9 | 2010, Proceedings of the 3rd International ICST Conference on Mobile Wireless Middleware, Operating Systems, and Applications | Using Smartphones to Detect Car Accidents and Provide Situational Awareness to Emergency Responders – Thompson, C., White, J., Dougherty, B., Albright, A., & Schmidt, D. C. | Smartphone sensors, GPS | Portable, low cost | Battery drain, low accuracy |
| 10 | 2019, International Journal of Engineering Research & Technology | Accident Detection and Emergency Notification System Using IoT – Gaikwad, V., & Patil, A. | IoT sensors, emergency alerts | Faster response, automation | Connectivity issues |

# CHAPTER 03: SOFTWARE REQUIREMENTS SPECIFICATIONS

## 3.1 Assumptions And Dependencies

- **Continuous Power Supply:** The system is assumed to receive uninterrupted power from the vehicle battery or external source.
- **Reliable GSM Connectivity:** The GSM module requires stable cellular network coverage for sending emergency SMS alerts.
- **GPS Signal Availability:** The GPS module should have proper satellite visibility for accurate real-time location tracking.
- **Proper Sensor Calibration:** Accelerometer, vibration, and flame sensors must be correctly calibrated to reduce false alerts.
- **Pre-Configured Emergency Contacts:** Emergency numbers must be stored in the system before operation.
- **Network Provider Support:** SMS delivery depends on the availability and quality of the GSM service provider.
- **Sensor Accuracy:** Accident detection accuracy depends on proper sensor sensitivity and threshold settings.

## 3.2 Functional Requirements

1. **Emergency Detection:** Ability to detect road accidents, fire hazards, and emergency situations using accelerometer, flame sensor, and emergency switch.
2. **Real-Time Location Tracking:** Automatic capture of GPS coordinates for accurate incident location identification.
3. **Automatic Alert Transmission:** Send emergency SMS alerts containing location and incident details to registered contacts and authorities.
4. **Manual Emergency Trigger:** Provide a push button for manually activating emergency alerts during critical situations.

5. **Alarm Activation:** Generate buzzer/alarm sound for immediate nearby attention and assistance.
6. **Incident Data Storage:** Store timestamp, GPS location, and sensor readings in local or cloud storage for analysis.
7. **Message Delivery Confirmation:** Ensure successful transmission and delivery of emergency notifications.
8. **Configuration Management:** Allow updating of emergency contacts, alert messages, and system settings.
9. **Autonomous Operation:** Automatic system functioning without continuous human intervention once powered on.
10. **Power Backup Support:** Maintain uninterrupted operation during power failures or emergency conditions.
11. **Automatic/Manual Reset:** Reset the system automatically or through user input after the emergency is resolved.
12. **Multi-Module Integration:** Integrate GPS, GSM, sensors, microcontroller, and IoT modules for coordinated operation.

## 3.3 External Interface Requirements

### 3.3.1 Hardware Interfaces

1. **GPS Interface:**

   The GPS module communicates with the microcontroller through UART serial communication for real-time location tracking.

2. **GSM Interface:**

   The GSM module interfaces with the controller using serial communication to transmit emergency SMS alerts.

3. **Sensor Interface:**

   Accelerometer, vibration, and flame sensors provide analog/digital input signals to the microcontroller for accident and fire detection.

4. **Buzzer Interface:**

   The buzzer is connected through digital output pins to generate alarm alerts during emergencies.

5. **Power Supply Interface:**

   The system operates using a regulated 5V power supply or vehicle battery connection.

### 3.3.2 Software Interfaces

1. **Operating System:** Windows 10 / Windows 11
2. **Development Environment:** Arduino IDE
3. **Programming Language:** Embedded C / C++
4. **Libraries Used:** TinyGPS++, SoftwareSerial, GSM libraries

### 3.3.3 Communication Interfaces

The communication interfaces enable data transfer between sensors, microcontroller, GPS, GSM module, and mobile devices for real-time emergency alert generation.

- **GPS Communication:** Serial communication for live location tracking
- **GSM Communication:** SMS transmission to emergency contacts and authorities
- **Sensor Communication:** Analog and digital communication between sensors and microcontroller
- **Alert Interface:** Emergency notifications through SMS and buzzer alarm

### 3.4 Non-Functional Requirements

- **Reliability:** The system should accurately detect accidents and fire hazards with minimal false alerts.
- **Scalability:** The system should support additional sensors or modules in future enhancements.
- **Efficiency:** The system should process sensor data and send alerts with minimum delay.
- **Maintainability:** The system design and code should support easy updates and maintenance.
- **Usability:** The system should operate automatically with minimal user interaction.
- **Availability:** The system should function continuously during vehicle operation and emergency conditions.

#### 3.4.1 Performance Requirements

- **Real-Time Detection:** The system must detect accidents or fire events immediately after sensor activation.
- **Fast Alert Generation:** Emergency SMS alerts should be sent within a few seconds of detection.
- **Location Accuracy:** GPS coordinates should provide accurate real-time incident location.
- **Continuous Monitoring:** Sensors should continuously monitor vehicle conditions without interruption.
- **Low False Detection:** The system should minimize false accident or fire alerts.

### 3.4.2 Safety Requirements

- **Emergency Notification:** The system must automatically notify emergency contacts during accidents or fire incidents.
- **Alarm Activation:** A buzzer should activate immediately during emergency detection.
- **Power Backup:** The system should support uninterrupted operation during power fluctuations.
- **Manual Emergency Support:** Users should be able to manually trigger alerts using an emergency switch.
- **Safe Operation:** Electrical components and sensors should operate safely under vehicle conditions.

### 3.4.3 Security Requirements

- **Authorized Access:** Only authorized users should modify emergency contact details or system settings.
- **Secure Communication:** Emergency messages and GPS data should be transmitted securely.
- **Data Protection:** Stored incident records should be protected from unauthorized access.
- **System Monitoring:** The system should maintain logs of detected incidents and alerts sent.

### 3.4.4 Software Quality Attributes

- **Portability:** The system software should run on Windows-based systems supporting Arduino IDE.

- **Robustness:** The system should function properly under varying environmental conditions and vibrations.
- **Interoperability:** The system should support integration with IoT platforms and additional safety modules.
- **Modifiability:** The software should allow future feature additions and sensor integration.
- **Flexibility:** The system should support updates in alert messages, contact numbers, and sensor thresholds.

### 3.5 System Requirements

- **Continuous Monitoring:** The system should continuously monitor vehicle conditions in real time.
- **Automatic Operation:** The system should work automatically without manual intervention after startup.
- **Multi-Module Integration:** GPS, GSM, sensors, and buzzer modules should operate together efficiently.
- **Alert Management:** The system should send alerts and maintain emergency notification records.
- **Modular Design:** Hardware and software modules should support independent maintenance and upgrades.

#### 3.5.1 Database Requirements

- **Incident Logs:** Store accident timestamps, GPS coordinates, and sensor readings.
- **Alert Records:** Maintain records of emergency SMS alerts sent to contacts.

- **Sensor Data Storage:** Store accelerometer, vibration, and flame sensor values for analysis.
- **Contact Information:** Store pre-configured emergency contact details securely.

### 3.5.2 Software Requirements (Platform Choice)

- **Operating System:** Windows 10 / Windows 11
- **Programming Language:** Embedded C / C++
- **IDE:** Arduino IDE
- **Libraries:** TinyGPS++, SoftwareSerial, GSM libraries

### 3.5.3 Hardware Requirements

- **Microcontroller:** ESP32 / Arduino UNO
- **GPS Module:** Neo-6M GPS Module
- **GSM Module:** SIM800L / SIM900A
- **Sensors:** Accelerometer, Vibration Sensor, Flame Sensor
- **Alarm Device:** Buzzer
- **Processor:** Intel i5 or higher for development and monitoring
- **RAM:** Minimum 4 GB RAM
- **Storage:** Minimum 20 GB free storage for Arduino IDE and project files
- **Power Supply:** 5V regulated supply or vehicle battery connection

### 3.6 Analysis Model: Iteration Model

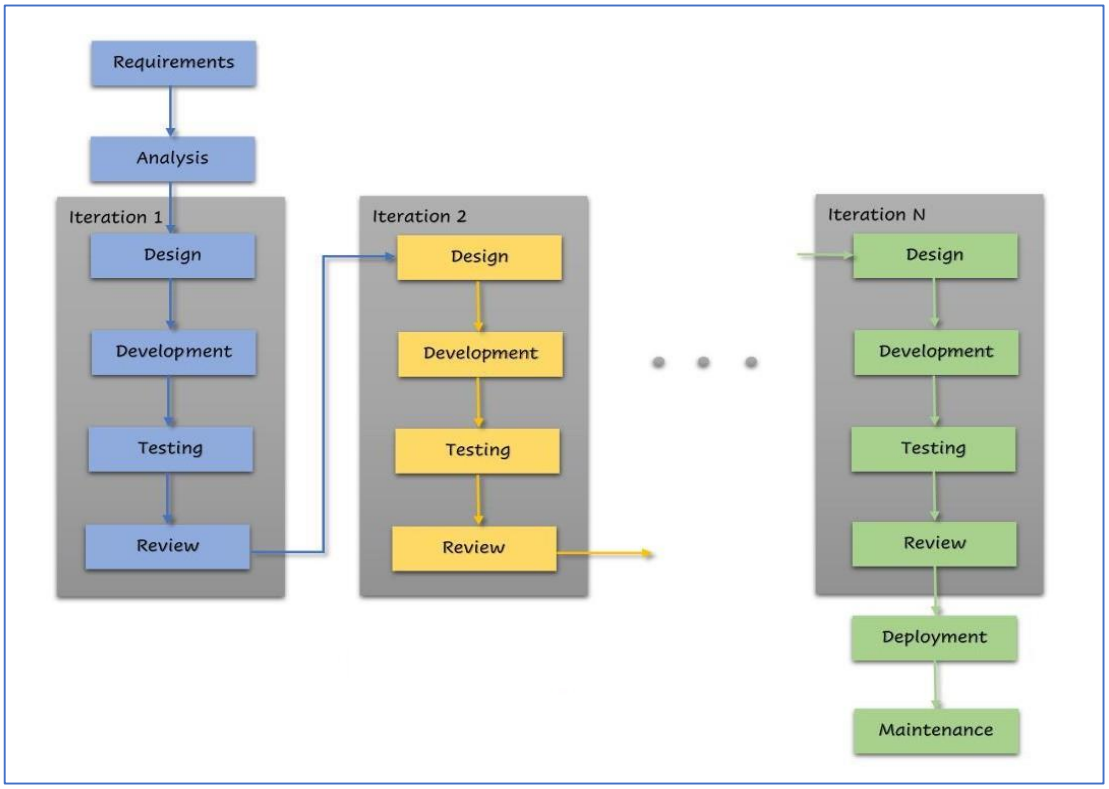



Fig 3.6.1 : Iterative Waterfall Model

The proposed IoT-Based Smart Accident Detection and Emergency Alert System follows the Iterative Waterfall Model. This model supports step-by-step development with continuous improvements after each iteration. It is suitable for the project because it allows gradual integration and testing of hardware and software modules.

**Iterative Waterfall Model Stages:**

➢ **Requirements:** Identified system objectives, sensors, GPS-GSM communication, and alert requirements.

➢ **Analysis:** Analysed accident detection methods, hardware feasibility, and system workflow.

➢ **Design:** Designed circuit connections, module interaction, and overall system architecture.

➢ **Development:** Implemented the system using Arduino IDE and Embedded C/C++.

➢ **Testing:** Tested sensors, GPS tracking, GSM alerts, and emergency detection functionality.

➢ **Review:** Improved system performance based on testing results and observations.

➢ **Deployment:** Integrated and executed the complete system on hardware modules.

➢ **Maintenance:** Performed debugging, calibration, and future feature enhancements.

# CHAPTER 04: PROJECT PLAN

## 4.1 Project Estimate

### 4.1.1 Reconciled Estimates:

| Sr. No | Parameter | Estimation |
|---|---|---|
| 1. | Project Type | Final Year IoT-Based Embedded System |
| 2. | Development Complexity | Moderate |
| 3. | Hardware Requirement | GPS, GSM, Sensors, Buzzer, Arduino/ESP32 |
| 4. | Hardware Cost | Low Cost (Approx. Rs.4000) |
| 5. | Software Requirement | Open-Source Tools and Libraries |
| 6. | Testing Requirement | Real-Time Sensor and Alert Testing |
| 7. | Power Requirement | 5V Regulated Supply / Vehicle Battery |
| 8. | Maintenance Requirement | Low |
| 9. | Scalability | Supports future sensor/module integration |

### 4.1.2 Project Resources:

**1. Hardware Components**

The proposed system requires different hardware modules for accident detection, communication, and emergency alert generation.

- **Arduino UNO / ESP32:** Main microcontroller for processing sensor data and controlling system operations.
- **GPS Module (Neo-6M):** Used for obtaining real-time location coordinates during emergencies.
- **GSM Module (SIM800L / SIM900A):** Used for sending emergency SMS alerts to registered contacts.
- **Accelerometer / Vibration Sensor:** Detects sudden impacts and accident conditions.
- **Flame Sensor:** Detects fire-related emergencies.
- **Buzzer:** Generates alarm alerts during emergency situations.

**2. Programming Libraries & Tools**

- **Arduino IDE:** Used for writing, compiling, and uploading Embedded C/C++ code to the microcontroller.
- **TinyGPS++ Library:** Used for GPS data processing and coordinate extraction.
- **SoftwareSerial Library:** Enables serial communication between GSM/GPS modules and the controller.
- **GSM Libraries:** Used for SMS transmission and communication handling.

**3. Supporting Software & Environment**

- **Windows 10 / Windows 11:** Operating system used for development and testing.
- **Serial Monitor & Serial Plotter:** Used for monitoring sensor values, GPS data, and system outputs during testing.
- **Embedded C / C++:** Programming language used for system implementation.

**4. Communication Resources**

- **GSM Network Connectivity:** Required for sending emergency SMS alerts.
- **GPS Satellite Connectivity:** Required for accurate real-time location tracking.
- **Power Supply / Vehicle Battery:** Provides continuous power to the system during operation.

## 4.2 Risk Management

### 4.2.1 Risk Identification

During the development of the IoT-Based Smart Accident Detection and Emergency Alert System, several risks were identified that could affect system performance and reliability. One major risk is false accident detection, where sudden vibrations, road bumps, or sensor fluctuations may trigger unnecessary emergency alerts. Similarly, false negatives may occur if an actual accident is not detected due to improper sensor calibration or hardware failure.

Another important risk is GSM network failure, where emergency SMS alerts may not be delivered because of poor cellular connectivity or network unavailability. GPS

signal loss is also a possible issue, especially in tunnels or densely covered areas, which can affect accurate location tracking.

Hardware-related risks such as sensor malfunction, loose connections, or power supply interruptions may reduce system reliability. Environmental conditions like excessive heat, dust, or vibration can also affect sensor performance.

Additionally, there is a risk of delayed emergency response if alert messages are not received on time by emergency contacts or authorities.

### 4.2.2 Risk Analysis

The identified risks vary in severity and impact on the system. False accident detection may reduce system reliability by generating unnecessary alerts, while failure to detect actual accidents has a higher level of risk because emergency assistance may not be provided on time.

Network-related issues such as GSM failure or weak GPS signals are highly possible in remote or low-connectivity areas and may affect message delivery and location accuracy. Hardware failures including faulty sensors or unstable power supply can interrupt system operation and reduce detection efficiency.

Environmental factors such as vibrations, weather conditions, and sensor sensitivity may also influence system accuracy. Proper testing, calibration, and regular maintenance are necessary to minimize these risks and improve overall system reliability.

### 4.2.3 Overview of Risk Mitigation, Monitoring, and Management

To reduce false accident detection, proper sensor calibration and threshold settings are used to differentiate actual accidents from normal vibrations or road disturbances. Multiple sensors such as accelerometer, vibration, and flame sensors are integrated to improve detection accuracy and reliability.

Hardware-related risks are minimized through regular testing of sensors, GSM, GPS, and power connections. Stable power supply support and proper circuit insulation help ensure continuous system operation during emergencies.

Network-related risks are managed by verifying GSM signal availability and testing SMS delivery regularly. GPS accuracy is improved by placing the module in locations with better satellite visibility. In case of temporary network failure, the system can retry message transmission once connectivity is restored.

Environmental risks such as heat, dust, and vibration are reduced by using properly mounted and protected hardware components. Continuous monitoring, real-time testing, and periodic maintenance help improve overall system performance and reliability.

### 4.3 Project Schedule

#### 4.3.1 Project Task Set

This section lists the major tasks performed during the development of the IoT-Based Smart Accident Detection and Emergency Alert System.

A. Planning & Analysis

- Define project scope and objectives
- Study existing accident detection systems
- Perform feasibility and risk analysis
- Identify hardware and software requirements

B. Hardware Selection & Setup

- Select microcontroller, GSM, GPS, and sensor modules
- Design circuit connections and hardware layout
- Configure power supply and communication interfaces

C. System Development

- Develop Embedded C/C++ programs using Arduino IDE
- Integrate accelerometer, vibration, flame, GPS, and GSM modules
- Implement emergency alert and buzzer functionalities

D. Communication & Alert Integration

- Configure GSM module for SMS transmission
- Integrate GPS location tracking with emergency alerts
- Store and verify emergency contact information

E. Testing & Evaluation

- Test accident and fire detection accuracy
- Verify SMS alert delivery and GPS coordinates
- Perform real-time testing under different conditions
- Calibrate sensors to reduce false alerts

F. Deployment

- Deploy the system on the complete hardware setup
- Connect modules for continuous real-time monitoring
- Validate overall system performance

G. Finalization

- Prepare project report and documentation
- Create circuit diagrams, flowcharts, and result analysis
- Conduct final project demonstration and presentation

### 4.3.2 Task Network

The task network represents the logical sequence and dependency between project activities during the development of the IoT-Based Smart Accident Detection and Emergency Alert System.

1. Planning & Analysis was the initial phase involving requirement gathering, feasibility study, and hardware/software selection.
2. After planning, Hardware Selection & Setup and System Development were carried out for sensor integration and circuit design.

3. Communication & Alert Integration depended on successful integration of GSM, GPS, and sensor modules.
4. Testing & Evaluation was performed after system implementation to verify accident detection, GPS tracking, and SMS alert generation.
5. Deployment was carried out after successful hardware and software testing.
6. Finalization included report preparation, documentation, and final project demonstration after deployment and evaluation.

### 4.3.3 Timeline Chart (Semester 1)

| **Activity** | **Week 1** | **Week 2** | **Week 3** | **Week 4** | **Week 5** | **Week 6** | **Week 7** | **Week 8** | **Week 9** | **Week 10** |
|---|---|---|---|---|---|---|---|---|---|---|
| | Aug 16 | Aug 23 | Aug 30 | Sep 6 | Sep 13 | Sep 20 | Sep 27 | Oct 4 | Oct 11 | Nov 1 |
| Topic Finalization & Planning | | | | | | | | | | |
| Requirement Gathering | | | | | | | | | | |
| Literature Survey | | | | | | | | | | |
| Feasibility Study | | | | | | | | | | |
| Hardware & Software Selection | | | | | | | | | | |
| System Design & Flow | | | | | | | | | | |
| Circuit Design & Setup | | | | | | | | | | |
| Module Testing | | | | | | | | | | |
| Research Paper Publication | | | | | | | | | | |

**Timeline Chart (Semester 2)**

| **Activity** | **Week 1** | **Week 2** | **Week 3** | **Week 4** | **Week 5** | **Week 6** | **Week 7** | **Week 8** | **Week 9** | **Week 10** |
|---|---|---|---|---|---|---|---|---|---|---|
| | Jan 17 | Jan 24 | Jan 31 | Feb 9 | Feb18 | Mar 5 | Mar18 | Apr13 | Apr20 | Apr27 |
| Hardware Integration | | | | | | | | | | |
| Detection Logic Implementation | | | | | | | | | | |
| GSM Alert System | | | | | | | | | | |
| GPS Tracking Integration | | | | | | | | | | |
| Emergency Alert Testing | | | | | | | | | | |
| Real-Time System Testing | | | | | | | | | | |
| Research Work Presentation in Conference | | | | | | | | | | |
| Sensor & SMS Validation | | | | | | | | | | |
| Research Paper Publication | | | | | | | | | | |
| Final Documentation & Black Book | | | | | | | | | | |

## 4.4 Team Organization

### 4.4.1 Team Structure

The project team consists of three members working collaboratively on the development of the IoT-Based Smart Accident Detection and Emergency Alert System. Responsibilities were distributed among the members based on project requirements and technical activities to ensure efficient project execution.

**1. Project Coordination & System Development**

- Managed project planning, task distribution, and coordination among team members.
- Designed the overall system architecture and handled hardware integration.
- Implemented Embedded C/C++ programming, sensor interfacing, GSM-GPS integration, testing, debugging, and system validation.
- Supervised documentation, report preparation, and presentation activities.

**2. Software & Communication Support**

- Assisted in software implementation using Arduino IDE and Embedded C/C++.
- Contributed to GSM communication, GPS tracking, and emergency alert functionalities.
- Supported testing, debugging, and performance improvement activities.

**3. Research & Documentation Support**

- Assisted in literature survey, research paper preparation, and documentation work.
- Helped in report formatting, presentation preparation, and data compilation.
- Supported project validation and testing observations.

### 4.4.2 Management Reporting and Communication

Effective communication and coordination were maintained throughout the project development process to ensure proper progress tracking and timely completion of tasks.

- Weekly discussions were conducted among team members to review project progress and technical issues.
- Regular guidance and feedback sessions were carried out with the project guide/faculty supervisor.
- Project tasks related to hardware, software, testing, and documentation were assigned and monitored systematically.

- Arduino IDE Serial Monitor outputs, testing observations, and debugging records were maintained during development.
- Project reports, source code, research paper documents, and presentation materials were shared among team members for collaboration.
- Final deliverables included the complete hardware setup, source code, research paper, technical documentation, and project presentation.

# CHAPTER 5: SYSTEM DESIGN

## 5.1 System Architecture

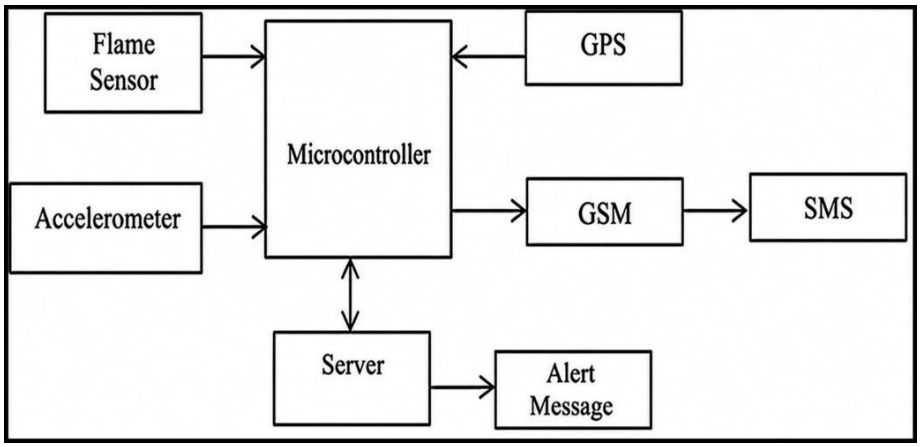



Fig 5.1.1 System Architecture

## 5.2 Mathematical Model

The Smart Emergency Response System can be mathematically modeled as a real-time event detection and alerting system based on sensor inputs, decision logic, and communication response. The model focuses on identifying emergency conditions and initiating automated alert actions.

### 1. System Definition

Let the system **S** be defined as:

$$S = \{I, P, O\}$$

where:

- **I** = Input set
- **P** = Processing and decision logic
- **O** = Output set

### 2. Input Set (I)

The input set consists of real-time sensor data collected from the vehicle environment:

$$I = \{A, G, F, L\}$$

where:

- **A** = Accelerometer readings (linear acceleration)

- **G** = Gyroscope readings (angular orientation)
- **F** = Flame sensor output
- **L** = GPS location coordinates

### 3. Threshold Conditions

Emergency conditions are detected based on predefined threshold values:

**Accident condition:**

$$A > A_{th} \text{ or } G > G_{th}$$

**Fire condition:**

$$F > F_{th}$$

where:

- $A_{th}$, $G_{th}$, **and** $F_{th}$ are threshold limits for acceleration, angular change, and flame detection respectively.

### 4. Event Detection Function

Let **E** represent the emergency detection function:

$$E = \begin{cases} 1, & \text{if } A > A_{th} \vee G > G_{th} \vee F > F_{th} \\ 0, & \text{otherwise} \end{cases}$$

where:

- **E = 1** indicates an emergency event
- **E = 0** indicates normal conditions

### 5. Processing and Decision Logic (P)

The processing unit (ESP32) applies multi-sensor verification to reduce false detections:

$$P = f(A, G, F)$$

If the emergency condition is validated over consecutive sampling intervals, the system confirms the event and initiates the alert process.

### 6. Output Set (O)

The output set includes alert and response actions:

$$O = \{M, B\}$$

where:

- **M** = Emergency alert message sent via GSM
- **B** = Buzzer activation for local alert
- **L** = GPS coordinates
- **T** = Timestamp
- **E_t** = Type of emergency (Accident / Fire / Medical)

The alert message **M** is defined as:

$$M = \{L, T, E_t\}$$

### 7. System Response

If **E = 1**, the system triggers the response mechanism:

$$S \rightarrow O$$

After successful message transmission, the system resets to continuous monitoring mode.

### 8. Success Condition

The system operation is considered successful if:

$$M_{delivered} = 1$$

indicating that the emergency alert message has been successfully transmitted to predefined emergency contacts. This mathematical representation validates the logical flow of the proposed system and supports its real-time emergency response capability.

### 5.3 Data Flow Diagram

In Data Flow Diagram (DFD), we represent the flow of data in the SmartResQ system. The DFD explains how input data from sensors is processed by the system and converted into emergency alerts and live location tracking information.

- In DFD Level 0, the overall flow of the system is shown using Input, Process, and Output.
- In DFD Level 1, the internal processing such as sensor monitoring, accident/fire detection, GPS tracking, and alert transmission is represented.
- In DFD Level 2, the detailed working of the SmartResQ system including sensor data analysis, emergency confirmation, GPS location retrieval, and GSM-based notification process is explained.

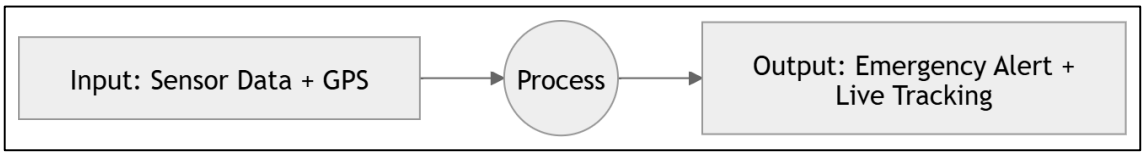



Fig 5.3.1: Data Flow(0) diagram

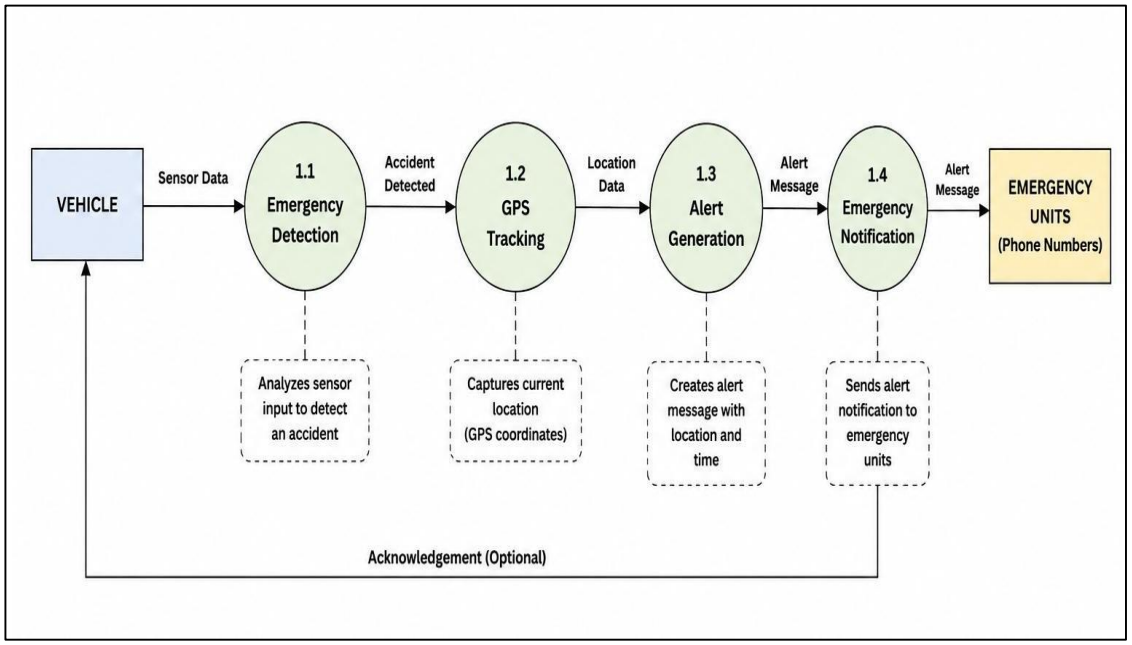



Fig 5.3.2: Data Flow(1) diagram

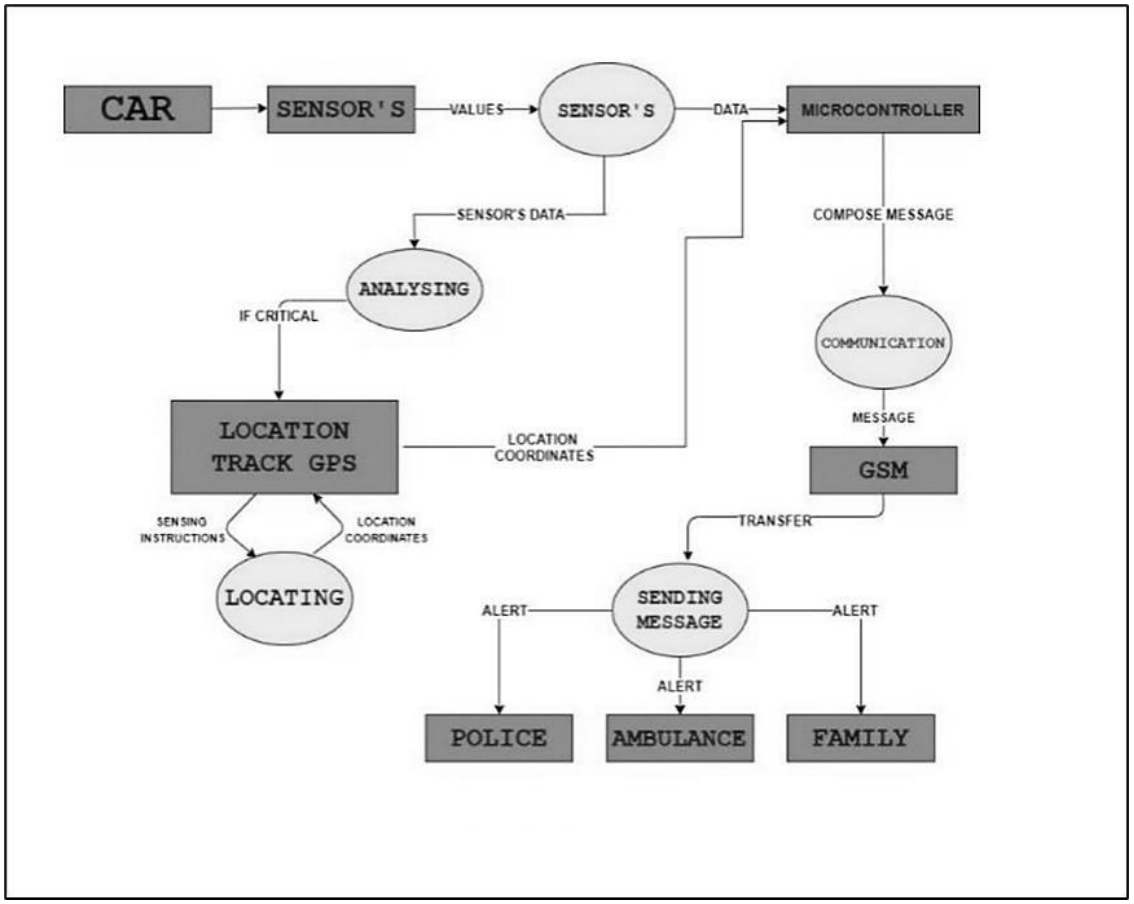



Fig 5.3.3: Data Flow(2) diagram

## 5.4 Entity Relationship Diagram (ERD)

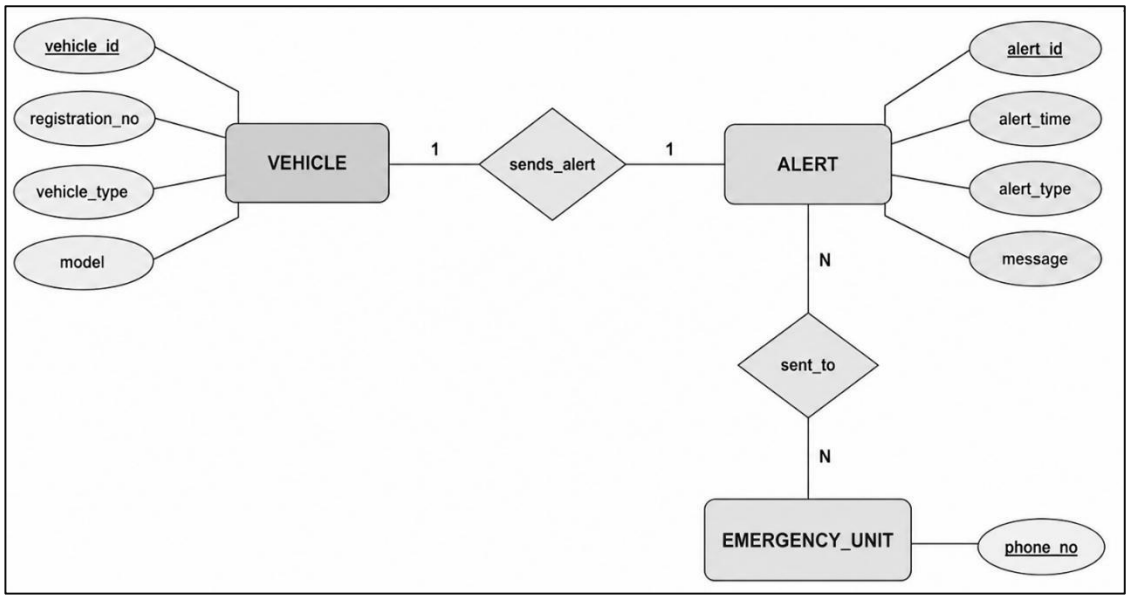



Fig 5.4 ER diagram

## 5.5 UML Diagrams

### 5.5.1 Activity Digram

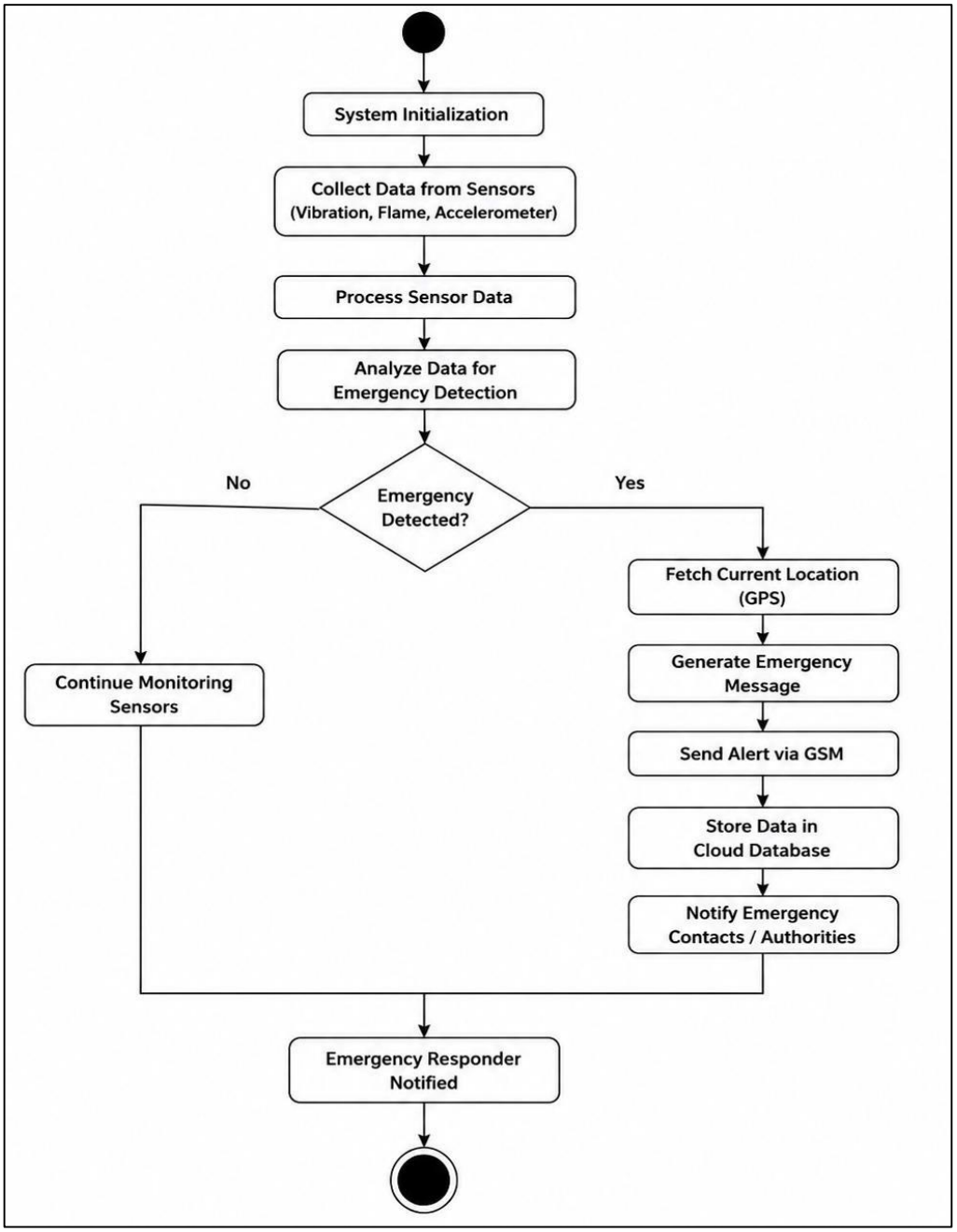



Fig 5.5.1 Activity Diagram

### 5.5.2 Class Diagram:

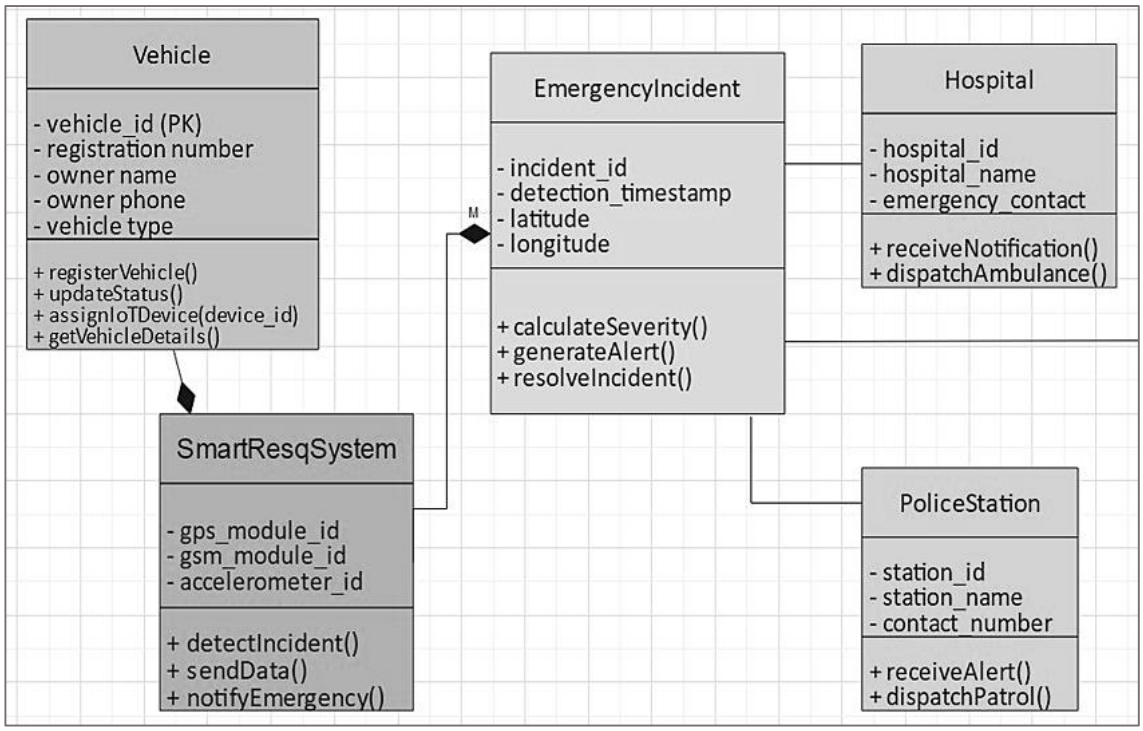



Fig 5.5.2 Class Diagram

### 5.5.3 Object Diagram:

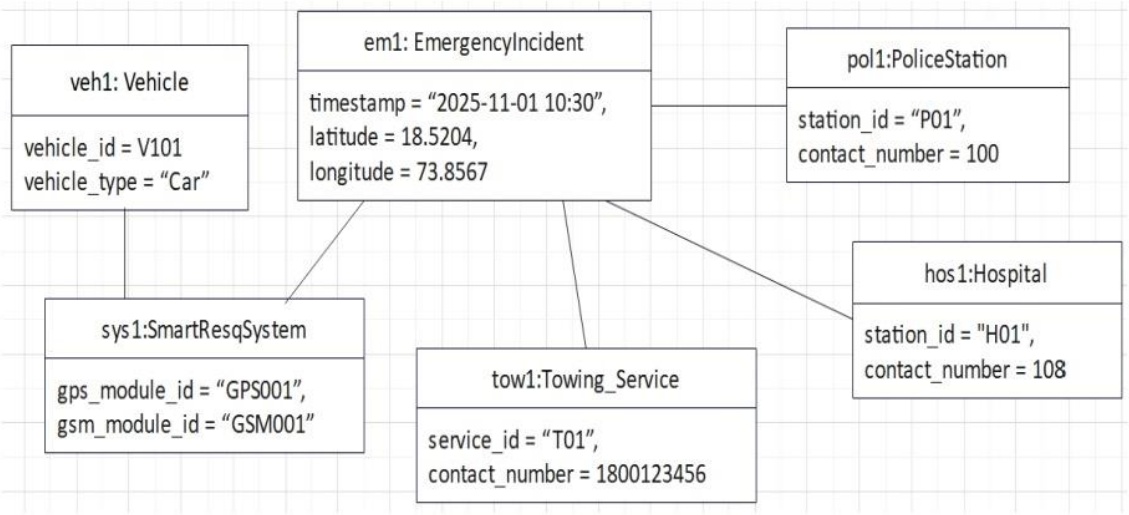



Fig. 5.5.3 Object Diagram

### 5.5.4 Usecase Diagram:

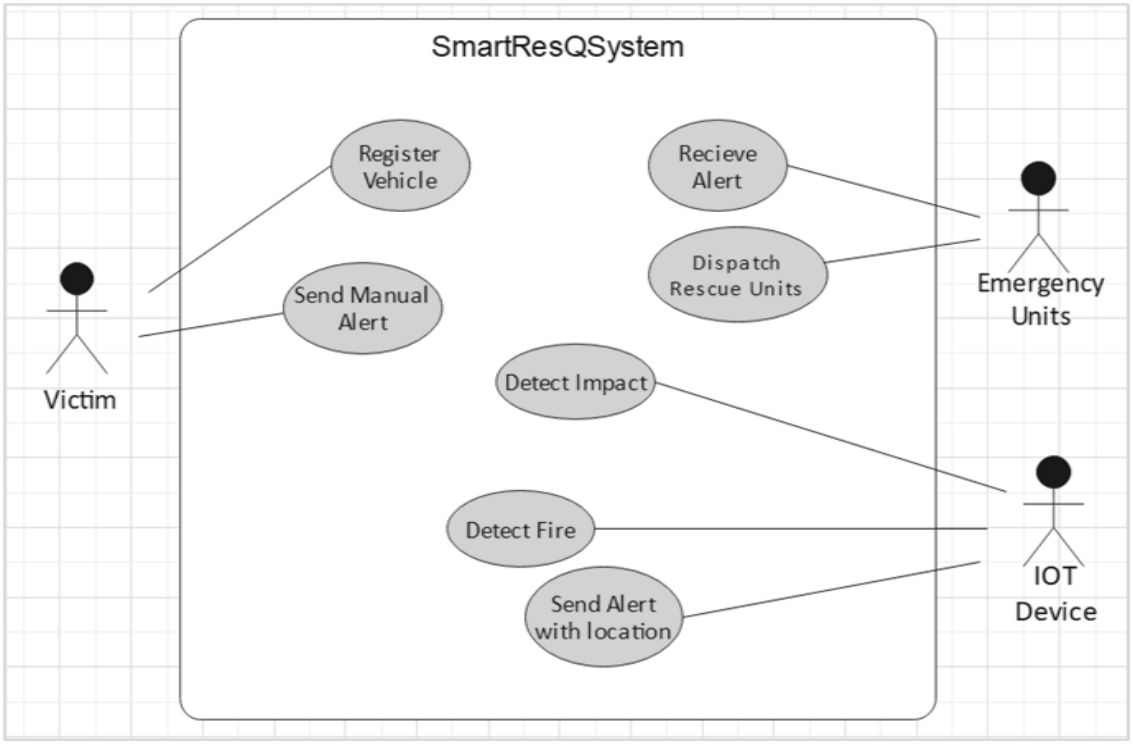



Fig 5.5.4 Usecase Diagram

### 5.5.5 Sequence Diagram:

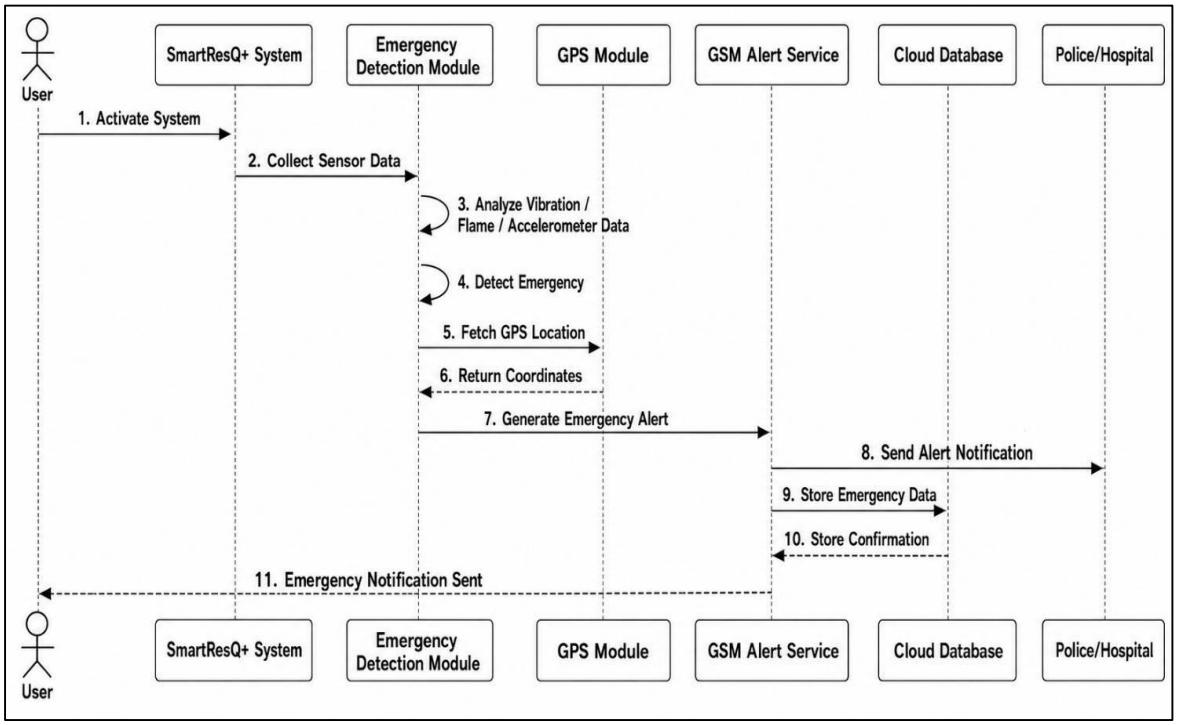



Fig 5.5.5 Sequence Diagram

### 5.5.6 State : System State Transitions:

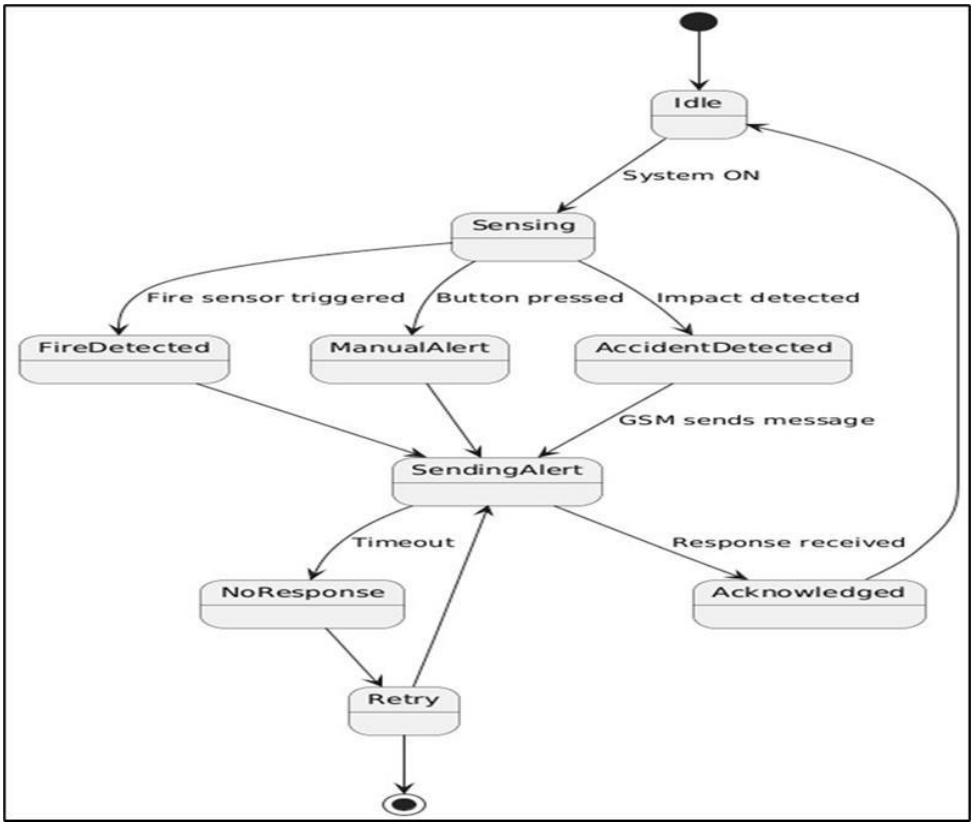



Fig 5.5.6 State Transition Diagram

### 5.5.7 Component Diagram:

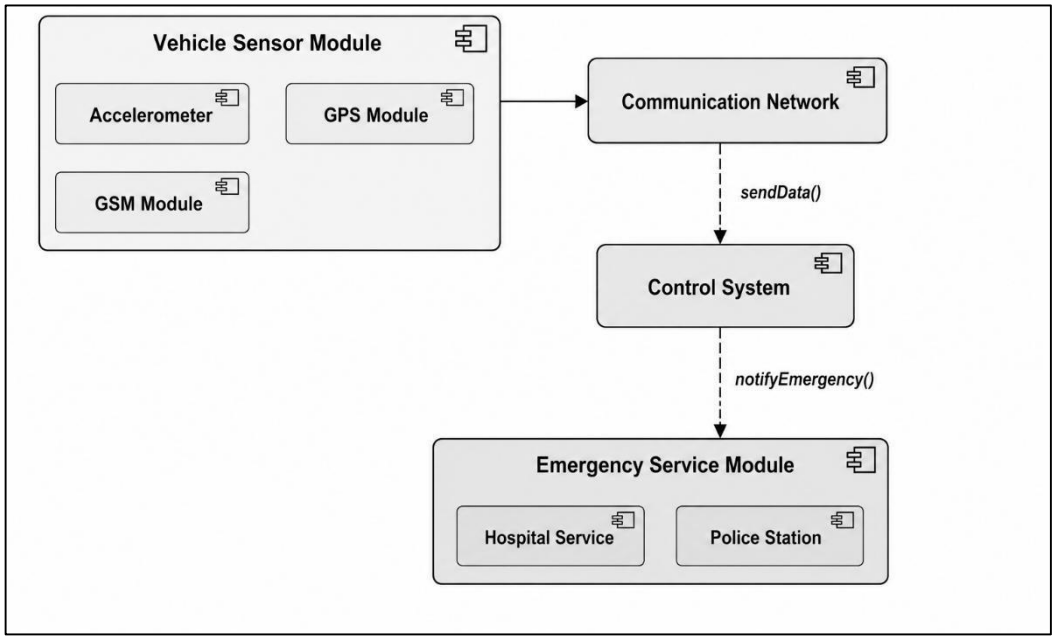



Fig. 5.5.7 Component Diagram

### 5.5.8 Deployment Diagram:

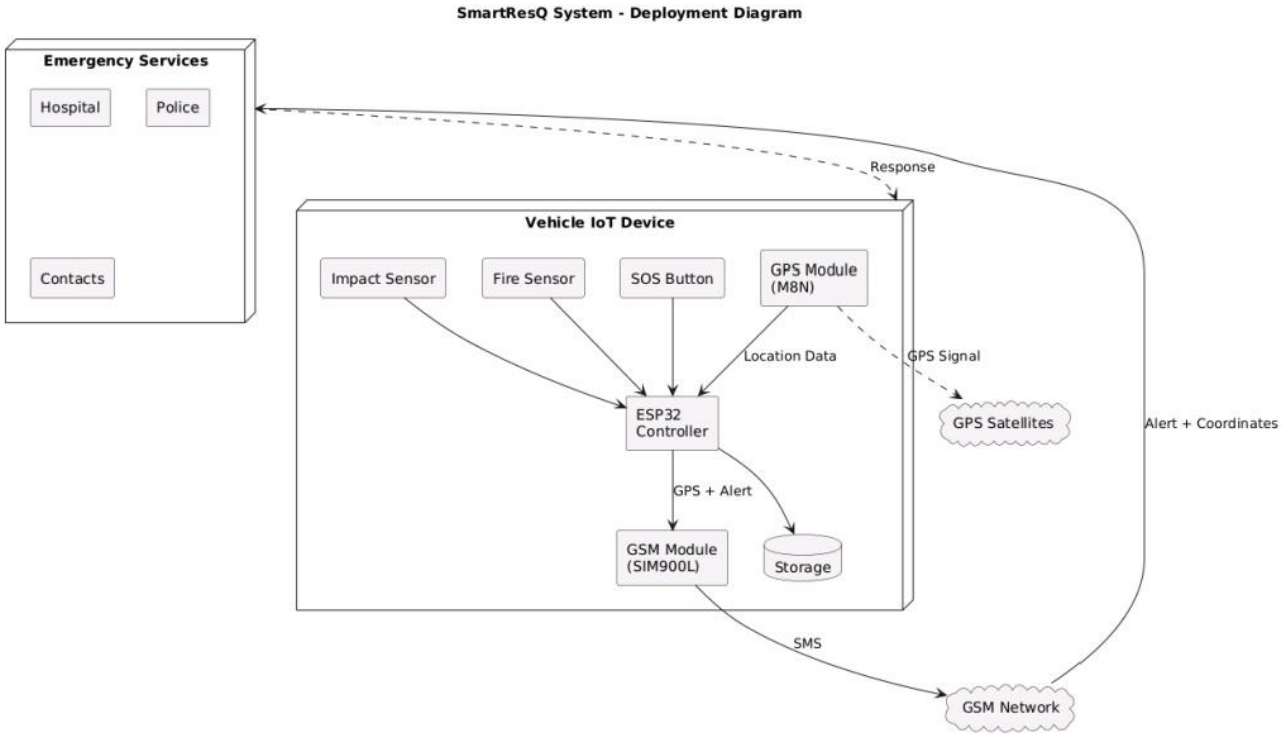



Fig. 5.5.8 Deployment Diagram

### 5.5.9 Package Diagram:

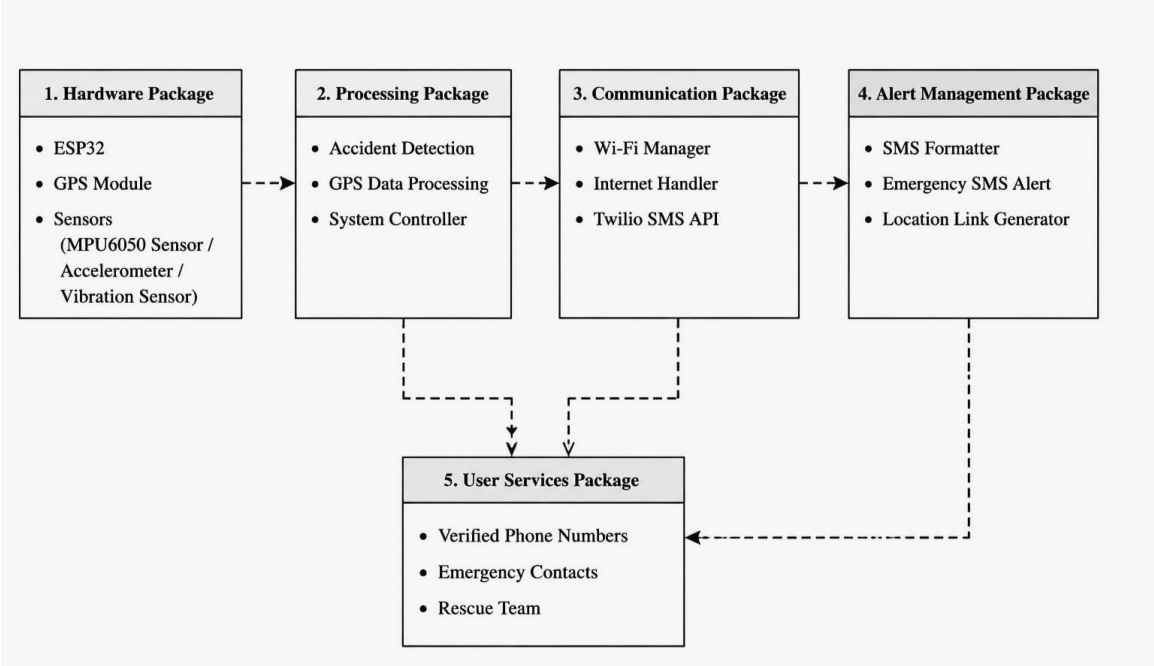



Fig. 5.5.9 Package Diagram

# CHAPTER 6: PROJECT IMPLEMENTATION

This section illustrates the process of developing, implementing, and testing the system. It explains the major modules, technologies, and algorithms involved in the Smart ResQ System.

## 6.1 Overview of Project Modules

The SmartResQ system consists of several functional modules that work together to detect emergencies, track location, and provide quick rescue response.

### 1. ESP32 Microcontroller Module

The ESP32 acts as the central processing unit of the system. It receives data from sensors, processes emergency conditions, and controls communication modules.

### 2. Sensor Monitoring Module

This module continuously collects real-time data from:

- Flame Sensor
- Accelerometer

The sensors monitor fire hazards, sudden impacts, abnormal motion, and accident conditions.

### 3. Emergency Detection Module

The collected sensor data is analyzed to detect emergency situations such as accidents or fire incidents based on predefined threshold values.

### 4. GPS Tracking Module

After emergency detection, the GPS module retrieves the real-time location coordinates of the affected user or vehicle.

### 5. GSM Alert Module

The GSM module generates and transmits emergency alert messages along with GPS location details to emergency contacts, hospitals, police stations, or rescue teams.

**6. Buzzer Alert Module**

The buzzer generates an audible warning signal during emergency situations to alert nearby people.

**7. Emergency Response Module**

This module enables rescue and emergency services to respond quickly based on the transmitted emergency alerts and location information.

### 6.2 Tools and Technologies Used

- Arduino IDE:
  Used for coding, compiling, and uploading programs to the ESP32 microcontroller.
- ESP32 Microcontroller:
  Acts as the main controller to process sensor inputs and manage system operations.
- GSM Module:
  Used to send emergency alert messages and notifications to emergency contacts, hospitals, and police authorities through the mobile network in Rural Areas.
- Flame Sensor:
  Detects fire or flame-related emergencies.
- Accelerometer Sensor:
  Detects sudden movement, tilt, or impact during accidents.
- Vibration Sensor:
  Detects vibrations and collision impact for accident identification.
- GPS Module:
  Provides real-time location tracking for emergency response and rescue operations.
- Buzzer:
  Generates emergency warning sounds and alerts during critical situations.
- Internet of Things (IoT):
  Enables communication between hardware components and emergency services in real time.

- Operating System: Windows
  Used for development, simulation, and system programming.

### 6.3 Algorithm Details

This sub-section outlines the fundamental algorithms employed within the project to detec Road  Emergency.

#### 6.3.1 Algorithm 1

**1. Sensor Data Collection**

The SmartResQ system continuously collects real-time data using the Flame Sensor and Accelerometer. The ESP32 microcontroller receives sensor readings and monitors sudden impacts, abnormal motion, and fire-related conditions. The collected sensor data is further forwarded for emergency analysis and processing.

**2. Emergency Detection**

After collecting sensor data, the system processes and analyzes the received values. The algorithm compares the sensor readings with predefined threshold values to identify emergency situations such as accidents or fire hazards. If abnormal conditions are detected, the system triggers the emergency alert mechanism.

**3. GPS Tracking**

Once an emergency is detected, the GPS module retrieves the real-time geographical coordinates of the affected user or vehicle. The system continuously updates the latitude and longitude information to ensure accurate location tracking during emergency situations.

**4. Alert Generation and GSM Transmission**

After obtaining the GPS location, the system generates an emergency alert message containing emergency details and live location information. The GSM module then transmits the alert message to emergency contacts, hospitals, police stations, and rescue authorities for immediate assistance.

**5. Buzzer Alert**

The buzzer module is activated immediately after emergency detection. It generates an audible warning signal to alert nearby individuals and create awareness about the emergency condition.

**6. Emergency Response**

After successful alert transmission, emergency services and rescue authorities receive the emergency information and location details. Based on the received alert, rescue operations and emergency response activities are initiated to provide quick assistance to the affected person.

# CHAPTER: 07: SOFTWARE TESTING

Testing is one of the most important phases in the development of the SmartResQ. Since the project is based on both hardware and software integration, various testing methods were carried out to verify the performance of individual components as well as the complete system.

## 7.1 Type of Testing

### 7.1.1 Unit Testing

Unit testing is the process of testing individual modules or components separately to verify whether they function correctly according to system requirements. In the SmartResQ project, unit testing was performed on each hardware module independently before integrating them into the complete system. The Accelometer MP6050 was tested to check its ability to detect sudden vibration or shock conditions. The GPS module was tested for accurate retrieval of location coordinates, while the GSM module was verified for proper message transmission functionality. Similarly, the buzzer and Arduino program logic were also tested individually to ensure correct operation. Unit testing helps in identifying faults at the initial stage of development and ensures that each module performs its intended task properly.

### 7.1.2 Integration Testing

Integration testing is performed after combining multiple modules into a single system. The purpose of this testing is to verify communication and interaction between different hardware components. In the SmartResQ system, integration testing was carried out after connecting the Accelometer MP6050 , GPS module, GSM module, and buzzer with the ESP32. The complete accident detection sequence was tested to ensure proper coordination among all modules. This testing verifies whether sensor data is processed correctly, GPS coordinates are retrieved successfully, and emergency messages are transmitted properly through the GSM module after accident detection.

### 7.1.3 Functional Testing

Functional testing is conducted to verify whether the system performs all required functions according to the project objectives. This testing focuses on validating the working functionality of the SmartResQ system.

The major functions tested during this phase include:

- Accident detection
- GPS location tracking
- Emergency SMS alert generation
- Alert buzzer activation
- Continuous monitoring of accident conditions

Functional testing ensures that all features of the system operate according to the specified requirements and produce the expected behavior during emergency situations.

### 7.1.4 Hardware Testing

Hardware testing is performed to verify the physical components and electronic circuits used in the project. This testing ensures that all hardware devices are connected properly and operate without electrical or communication issues.

The hardware components tested in the SmartResQ system include:

- ESP 32
- Viberation Sensor
- GPS module
- GSM module
- Buzzer
- Power supply unit
- Connecting wires and breadboard/PCB
- Flame sensor

During hardware testing, parameters such as voltage supply, signal transmission, circuit stability, and sensor responsiveness were examined carefully.

### 7.1.5 Software Testing

Software testing is carried out to verify the correctness of the Arduino program and embedded system logic used in the project. The objective of this testing is to ensure that the program executes properly without syntax errors, logical errors, or communication failures.

The software code was tested for:

- Sensor input processing
- Accident detection conditions
- Serial communication with GPS and GSM modules
- SMS transmission commands
- Timing and delay functions

Software testing ensures that the programmed instructions perform accurately and coordinate properly with the connected hardware modules.

### 7.1.6 Performance Testing

Performance testing is conducted to evaluate the speed, efficiency, and responsiveness of the SmartResQ system during accident situations. This testing measures how quickly the system reacts after detecting an accident.

The performance parameters considered during testing include:

- Accident detection response time
- GPS location retrieval time
- Emergency SMS transmission time
- System processing speed

Performance testing is important because quick emergency response plays a major role in reducing accident severity and improving rescue efficiency.

### 7.1.7 Reliability Testing

Reliability testing is performed to verify whether the system can operate continuously for a long period without failure. Since the SmartResQ system is designed for emergency applications, reliability is a critical requirement.

During this testing, the system was operated continuously under different conditions to monitor:

- Stability of hardware modules
- Consistency of accident detection
- Communication reliability
- Continuous power operation

Reliability testing helps in ensuring that the system remains dependable during real-time operation.

### 7.1.8 User Acceptance Testing

User Acceptance Testing (UAT) is conducted to verify whether the developed system satisfies the project requirements and fulfills the intended objectives. This testing is generally performed from the user's perspective to evaluate system usability and practical implementation.In the SmartResQ project, the complete system was tested in simulated accident conditions to examine its real-world applicability. The testing process focused on checking the ease of operation, emergency alert functionality, and overall system effectiveness.User Acceptance Testing ensures that the developed system is practical, efficient, and suitable for accident rescue applications.

### 7.2 Test Cases & Test Results

| Test Case ID | Test Description | Input Condition | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| 01 | System Power ON | Provide power supply | System should initialize successfully | System initialized correctly | Pass |
| 02 | Sensor Detection Test | Apply vibration/shock | Sensor should detect accident | Accident detected successfully | Pass |
| 03 | GPS Location Test | Enable GPS module | GPS should fetch coordinates | Accurate coordinates received | Pass |
| 04 | GSM SMS Test | Trigger accident condition | SMS should be sent | SMS sent successfully | Pass |
| 05 | Buzzer Alert Test | Accident occurrence | Buzzer should activate | Alert sound generated | Pass |
| 06 | Communication Test | Connect all modules | Modules should communicate properly | Communication successful | Pass |
| 07 | Continuous Operation Test | Run system continuously | System should work without interruption | Stable operation observed | Pass |
| 08 | False Alert Test | Normal movement condition | No emergency SMS should be sent | False alerts minimized | Pass |
| 09 | Power Stability Test | Variable power supply | System should remain operational | Stable performance observed | Pass |

# CHAPTER 08: RESULTS

## 8.1 Outcomes

The SmartResQ System produced several important outcomes related to emergency response, safety improvement, reliability, and system efficiency. The implementation of IoT technologies such as GPS, GSM, sensors, and ESP 32 helped in developing a smart and automated rescue solution capable of responding quickly during emergency situations. The major performance outcomes achieved during the project implementation are explained below.

### 1. Reduced Response Time

One of the most significant outcomes of the SmartResQ system is the reduction in emergency response time. Traditional accident reporting methods depend heavily on human involvement, which often causes delays in informing rescue teams or family members. In many cases, victims may not be able to call for help immediately after an accident due to injuries or unconscious conditions. The SmartResQ system overcomes this limitation by automatically detecting accidents and generating emergency alerts within a minimized time accuracy of 2-3 sec.

### 2. High Reliability and Accuracy

The system demonstrated high reliability and operational accuracy during testing and implementation. The SmartResQ system achieved nearly 95–97% accuracy in accident detection and GPS-based location tracking. The Accelometer MP6050 successfully identified accident-like conditions, while the GPS module provided accurate geographical coordinates of the accident location. The system also reduced false alerts by differentiating between normal vehicle movement and sudden collision conditions.

### 3. Multi-Scenario Emergency Handling

Another major outcome of the project is its ability to handle multiple emergency scenarios. Although primarily designed for accident detection, the SmartResQ system can also be adapted for other emergency situations such as:

- Medical emergencies
- Fire hazard conditions
- Sudden abnormal vehicle events

This multi-scenario capability increases the flexibility and practical usability of the system. The project can therefore be used not only in road safety applications but also in smart emergency monitoring systems for public transportation and safety management.

**4. Improved System Efficiency**

The SmartResQ system demonstrated efficient operation in terms of power consumption, processing speed, and communication reliability. The system continuously monitors accident conditions while consuming comparatively low electrical power, making it suitable for long-term vehicle installation. The communication between ESP32, GPS module, GSM module, and sensors remained stable throughout testing.

**5. Enhanced Safety and Automation**

The project successfully automated the process of accident monitoring and emergency alert generation. Unlike manual reporting systems, SmartResQ automatically detects accidents and informs emergency contacts without requiring human intervention. This automation improves road safety by ensuring immediate communication during emergencies.

**6. Cost-Effective and Scalable Solution**

The SmartResQ system was developed using affordable and easily available components such as ESP32, GSM module, GPS module, and sensors. This makes the project economical and suitable for practical implementation in vehicles. In addition, the project has future scalability. Advanced technologies such as cloud computing, mobile applications, AI-based monitoring, and live emergency dashboards can be integrated into the existing system for improved functionality and smart transportation applications.

## 8.2 Experimental Setup and Enclosure Design:

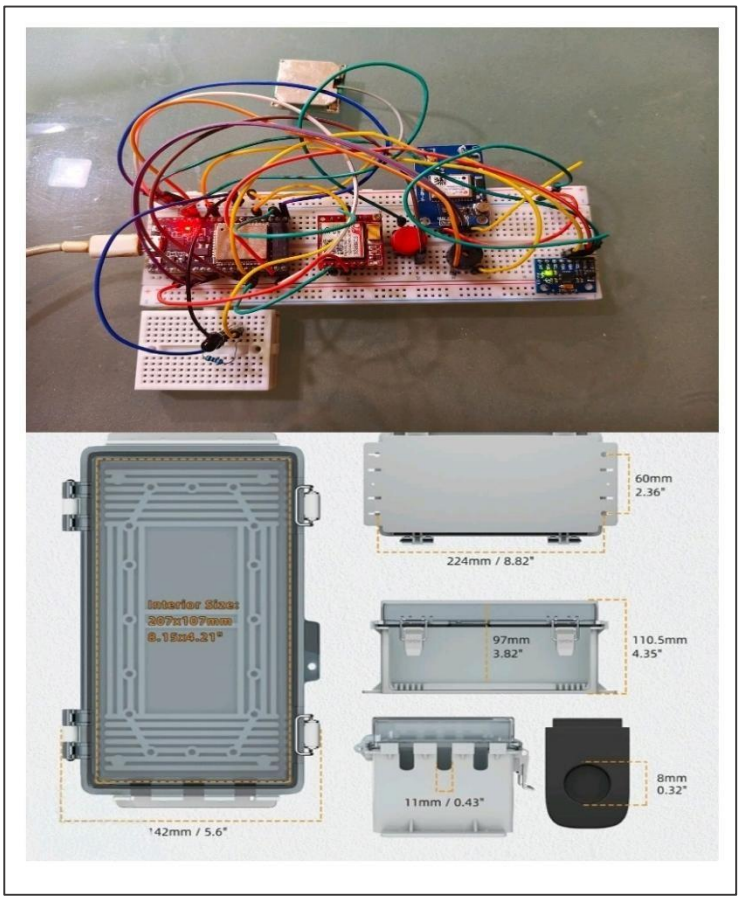
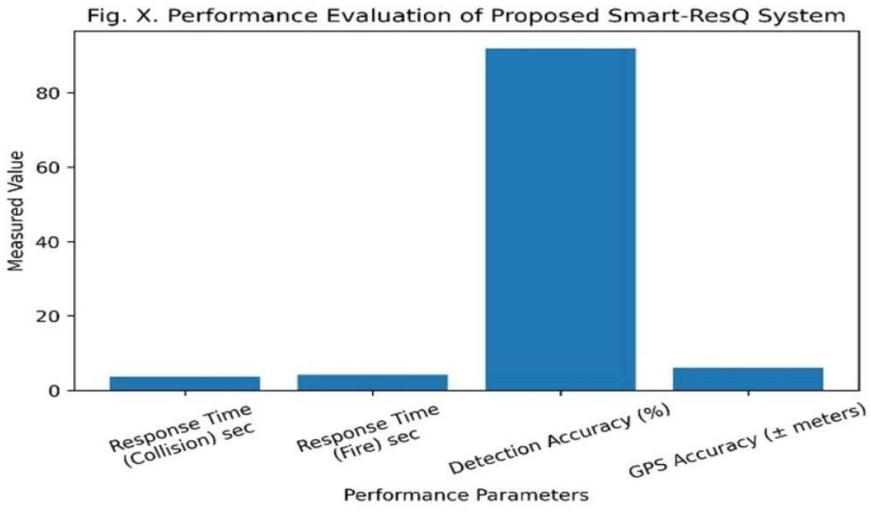
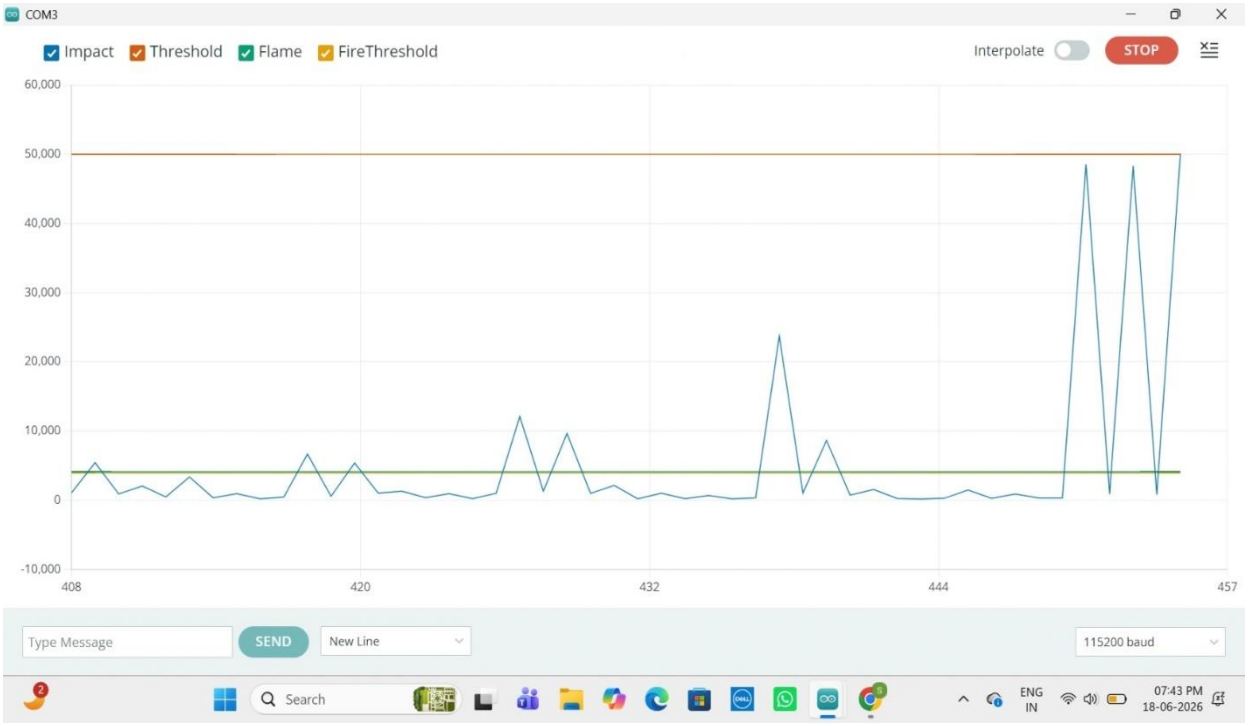
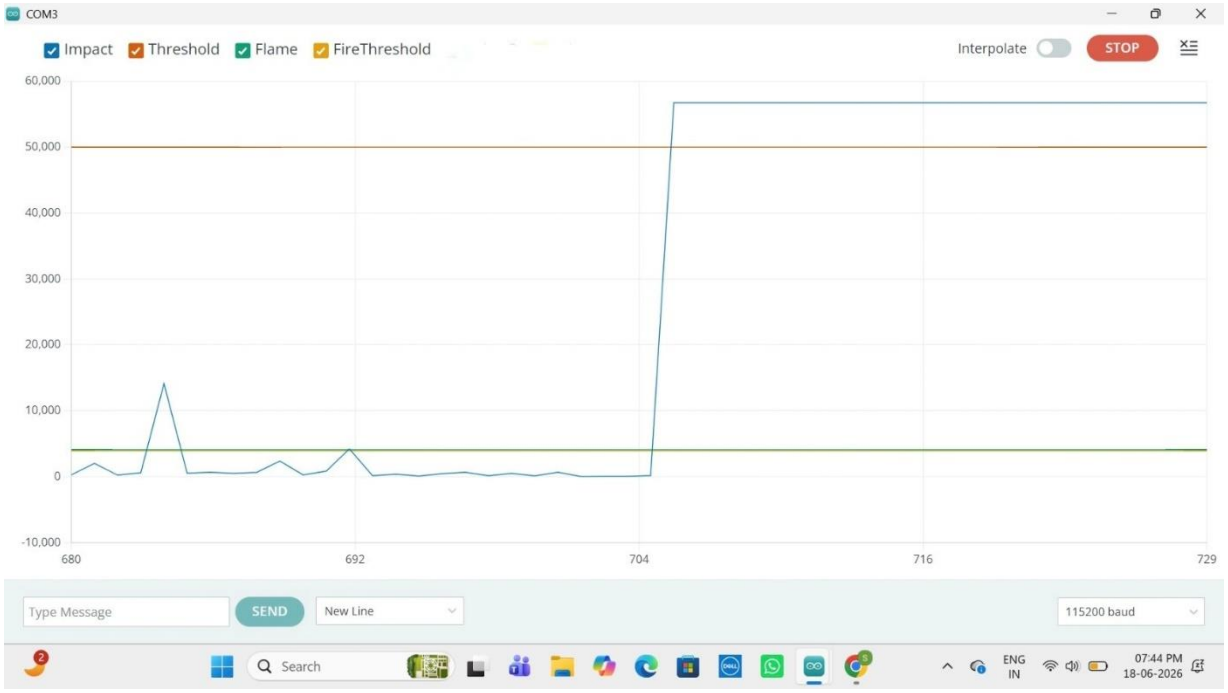



Fig 8.2.1 Experimental Setup and Enclosure Design of Proposed System



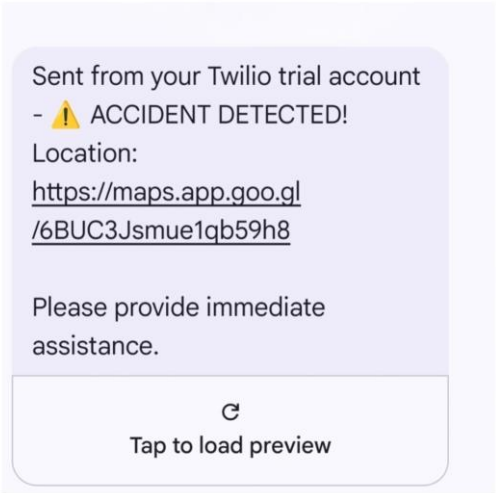
❖ **Threshold Graph:**

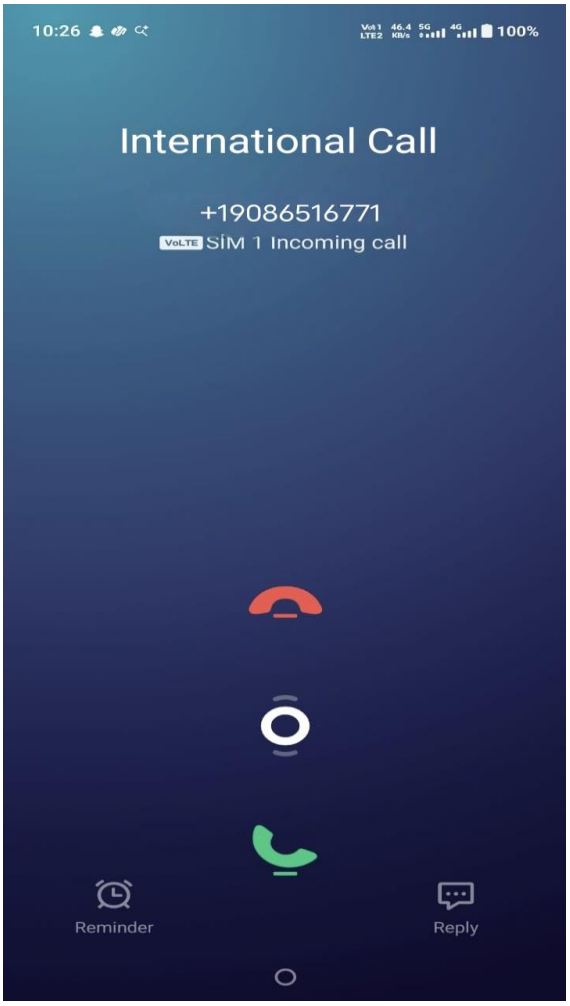


❖ **Received Message & Call :**



# CHAPTER 09: CONCLUSION

## 9.1 Conclusion

The SmartResQ System was successfully designed and implemented as an IoT-based emergency response solution for road safety applications. The system reduces delays in accident reporting and improves rescue response time using technologies such as ESP32, GPS module, GSM module, and MPU6050 accelerometer sensor.The developed system automatically detects accidents and immediately sends emergency alerts along with live location details to predefined contacts. This helps in providing faster medical assistance during critical situations and reduces dependence on manual communication.

The project successfully integrated hardware and software components into a reliable real-time monitoring system. During testing, the SmartResQ system demonstrated accurate accident detection, reliable GPS tracking, and successful SMS alert transmission. The system also showed low power consumption, quick response time, and efficient overall performance.

In conclusion, the SmartResQ System provides a smart, cost-effective, and reliable solution for emergency detection and rescue support. With future enhancements such as mobile application integration and cloud connectivity, the system can be further developed for large-scale real-world deployment.

## 9.2 Future Work

Although the SmartResQ system successfully achieved its objectives, several future improvements can further enhance its performance and usability.

• Cloud Computing Integration

The system can be integrated with cloud platforms for remote storage and monitoring of accident data. This would help emergency authorities access real-time information from any location.

• Mobile Application Development

A dedicated mobile application can be developed to provide live accident notifications, vehicle tracking, emergency contact management, and direct communication with hospitals and police stations.

• AI and Machine Learning Integration

Artificial Intelligence and Machine Learning algorithms can be implemented to reduce false alerts and improve accident detection accuracy. AI-based analysis can also help predict accident risks using sensor data and driving patterns.

• Smart City Implementation

SmartResQ can become a part of smart city infrastructure by providing real-time accident monitoring and reporting. Government authorities can use this information to improve traffic management and emergency response coordination.

• Centralized Government Monitoring Dashboard

A centralized dashboard can be developed for government authorities to monitor accident reports from multiple locations in real time. This will enable better decision-making and efficient allocation of emergency resources.

• Integration with Government Hospitals

The system can automatically notify nearby government hospitals when a severe accident is detected. This will help medical staff prepare in advance and reduce delays in providing treatment.

• Traffic Management Support

Accident alerts generated by SmartResQ can be shared with traffic control departments. This will help authorities manage traffic diversions, reduce congestion, and improve road safety during emergencies.

• National Road Safety Data Collection

The project can contribute to a national accident database maintained by the government. Such data can support research, policy-making, and the development of new road safety regulations.

## 9.3 Applications

The SmartResQ – IoT Based Accident Rescue System has various practical applications in transportation safety and emergency response systems.

1. **Automobile Safety Systems**
   The system can be installed in cars, buses, trucks, and motorcycles to automatically detect accidents and send emergency alerts, thereby improving passenger safety.
2. **Ambulance and Emergency Vehicles**
   The system can be used for real-time vehicle monitoring and tracking in ambulances and emergency vehicles to improve rescue coordination.
3. **Public Transportation Systems**
   SmartResQ can be implemented in buses and school vehicles to enhance passenger safety and provide automatic emergency reporting during accidents.
4. **Smart City Applications**
   The project can be integrated with smart city infrastructure and traffic management systems for better public safety and emergency response management.
5. **Highway and Traffic Safety Monitoring**
   The system can be deployed on highways and accident-prone areas for faster accident reporting and quick rescue operations, helping reduce fatalities caused by delayed medical assistance.

# APPENDIX A

A1. Problem statement feasibility assessment using, satisfiability analysis and NP Hard, NP-Complete or P type using modern algebra and relevant mathematical models.

The proposed project, “IoT-Based Smart Accident Detection and Fire Alert System,” is technically and operationally feasible using available IoT hardware components such as sensors, GSM, GPS modules, and microcontrollers. The system automatically detects accidents or fire emergencies and sends alert messages with location details to emergency contacts.

The feasibility of the system is verified using satisfiability analysis. The required conditions such as accident/fire detection, GPS activation, and successful SMS transmission can occur simultaneously, making the system logically satisfiable.

The system logic can be represented as:

$$(A + F) \cdot G \cdot S$$

Where:

- (A) = Accident detected
- (F) = Fire detected
- (G) = GPS active
- (S) = SMS sent successfully

The project mainly performs sensor monitoring, threshold checking, and alert generation, which are executed in polynomial time. Hence, the problem belongs to the P-Type category and is neither NP-Hard nor NP-Complete.

Boolean algebra and simple mathematical threshold models are used for system decision-making and feasibility analysis.

# APPENDIX B

## A) Journal Publication Details

**Paper Title:** Smart-ResQ – An IOT based Emergency Detection, Alerting and Resque System.

1. Name of the Journal where the paper was published:
   - International Journal For Multidisciplinary Research (IJFMR), Volume 07, Issue 06, November-December 2025.
   - DOI : https://doi.org/10.36948/ijfmr.2025.v07i06.60839
2. Paper Accepted/Rejected: Accepted and Published.
3. Review Comments by Reviewer: No comments.



International Journal For Multidisciplinary Research

An International Open Access Peer Reviewed Journal • Impact Factor: 9.24 • E-ISSN: 2582-2160

# Certificate of Publication

The editorial board of IJFMR is hereby awarding the certificate of publication to

**Dr. Madhuri Borawake**

in recognition of publication of the paper titled

**Smart-ResQ – An IoT-Based Emergency Detection, Alerting, and Rescue System**

Co-author(s): Ms. Sharvari .B. Chambhare, Ms. Neha Baliram Shinge, Ms. Sai Deshmukh

Published In: **Volume 7, Issue 6 (November-December 2025)**

Paper Id: **60839**

Editor / Publisher
IJFMR

www.ijfmr.com • editor@ijfmr.com



International Journal For Multidisciplinary Research

An International Open Access Peer Reviewed Journal • Impact Factor: 9.24 • E-ISSN: 2582-2160

Certificate of Publication

The editorial board of IJFMR is hereby awarding the certificate of publication to

**Ms. Sharvari .B. Chambhare**

in recognition of publication of the paper titled

**Smart-ResQ – An IoT-Based Emergency Detection, Alerting, and Rescue System**

Co-author(s): Ms. Neha Baliram Shinge, Ms. Sai Deshmukh, Dr. Madhuri Borawake

Published In: **Volume 7, Issue 6 (November-December 2025)**

Paper Id: **60839**

www.ijfmr.com • editor@ijfmr.com

Editor / Publisher
IJFMR



International Journal For Multidisciplinary Research

An International Open Access Peer Reviewed Journal • Impact Factor: 9.24 • E-ISSN: 2582-2160

Certificate of Publication

The editorial board of IJFMR is hereby awarding the certificate of publication to

**Ms. Neha Baliram Shinge**

in recognition of publication of the paper titled

**Smart-ResQ – An IoT-Based Emergency Detection, Alerting, and Rescue System**

Co-author(s): Ms. Sharvari .B. Chambhare, Ms. Sai Deshmukh, Dr. Madhuri Borawake

Published In: **Volume 7, Issue 6 (November-December 2025)**

Paper Id: **60839**

Editor / Publisher
IJFMR

www.ijfmr.com • editor@ijfmr.com



International Journal For Multidisciplinary Research

An International Open Access Peer Reviewed Journal • Impact Factor: 9.24 • E-ISSN: 2582-2160

# Certificate of Publication

The editorial board of IJFMR is hereby awarding the certificate of publication to

**Ms. Sai Deshmukh**

in recognition of publication of the paper titled

**Smart-ResQ – An IoT-Based Emergency Detection, Alerting, and Rescue System**

Co-author(s): Ms. Sharvari .B. Chambhare, Ms. Neha Baliram Shinge, Dr. Madhuri Borawake

Published In: **Volume 7, Issue 6 (November-December 2025)**

Paper Id: **60839**

Editor / Publisher
IJFMR

www.ijfmr.com • editor@ijfmr.com

**B) Conference Presentation Details**

Paper Title: Design and Implementation of an IoT-Based Smart Emergency Alert System

1. Name of the Conference where the paper was submitted/presented:
    - Second International Conference on Engineering, Science and Management (ICESM-2026), organized by Guru Gobind Singh College of Engineering and Research Centre, Nashik, Maharashtra, India in association with RSP Research Hub, Coimbatore, Tamil Nadu, India.
2. Paper Accepted/Rejected: Accepted and Presented.
3. Review Comments by Reviewer: No comments.



# CERTIFICATE

## of Participation

This Certificate is Proudly Presented to

**Dr. Madhuri P. Borawake**

HoD, Department of Computer Engineering, PDEA's College of Engineering, Pune, India

**For attending and giving an oral presentation on the paper entitled**

Design and Implementation of an IoT-Based Smart Emergency Alert System

at the **"Second International Conference on Engineering, Science and Management (ICESM-2026)"**- Jointly Organized by the Department of Computer Engineering & Electrical Engineering, Guru Gobind Singh College of Engineering and Research Centre, Nashik, Maharashtra, India & RSP Research Hub, Coimbatore, Tamil Nadu, India on **10.04.2026 & 11.04.2026.**

| **Dr. Nita Thakare** | **Dr. Rahul Agrawal** | **Dr. N.G. Nikam** |
|---|---|---|
| HoD, Computer Engineering (Convenor), Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. | HoD, Electrical Engineering (Convenor), Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. | Principal, Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. |

GGSF & RSP ICESM 2026 - 562



# CERTIFICATE

## of Participation

This Certificate is Proudly Presented to

**Ms. Sharvari Baban Chambhare**

Student, Department of Computer Engineering, PDEA's College of Engineering, Pune, India

**For attending and giving an oral presentation on the paper entitled**

Design and Implementation of an IoT-Based Smart Emergency Alert System

at the **"Second International Conference on Engineering, Science and Management (ICESM-2026)"**- Jointly Organized by the Department of Computer Engineering & Electrical Engineering, Guru Gobind Singh College of Engineering and Research Centre, Nashik, Maharashtra, India & RSP Research Hub, Coimbatore, Tamil Nadu, India on **10.04.2026 & 11.04.2026.**

| **Dr. Nita Thakare** | **Dr. Rahul Agrawal** | **Dr. N.G. Nikam** |
|---|---|---|
| HoD, Computer Engineering (Convenor), Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. | HoD, Electrical Engineering (Convenor), Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. | Principal, Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. |

GGSF & RSP ICESM 2026 - 563



# CERTIFICATE

## of Participation

This Certificate is Proudly Presented to

**Ms. Neha Baliram Shinge**

Student, Department of Computer Engineering, PDEA's College of Engineering, Pune, India

**For attending and giving an oral presentation on the paper entitled**

Design and Implementation of an IoT-Based Smart Emergency Alert System

at the **"Second International Conference on Engineering, Science and Management (ICESM-2026)"**- Jointly Organized by the Department of Computer Engineering & Electrical Engineering, Guru Gobind Singh College of Engineering and Research Centre, Nashik, Maharashtra, India & RSP Research Hub, Coimbatore, Tamil Nadu, India on **10.04.2026 & 11.04.2026.**

| Dr. Nita Thakare | Dr. Rahul Agrawal | Dr. N.G. Nikam |
|---|---|---|
| HoD, Computer Engineering (Convenor), Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. | HoD, Electrical Engineering (Convenor), Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. | Principal, Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. |

GGSF & RSP ICESM 2026 - 564



# CERTIFICATE

## of Participation

This Certificate is Proudly Presented to

**Ms. Sai Deshmukh**

Student, Department of Computer Engineering, Pdea's College of Engineering, Pune, India

**For attending and giving an oral presentation on the paper entitled**

Design and Implementation of an IoT-Based Smart Emergency Alert System

at the **"Second International Conference on Engineering, Science and Management (ICESM-2026)"**- Jointly Organized by the Department of Computer Engineering & Electrical Engineering, Guru Gobind Singh College of Engineering and Research Centre, Nashik, Maharashtra, India & RSP Research Hub, Coimbatore, Tamil Nadu, India on **10.04.2026 & 11.04.2026.**

| **Dr. Nita Thakare** | **Dr. Rahul Agrawal** | **Dr. N.G. Nikam** |
|---|---|---|
| HoD, Computer Engineering (Convenor), Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. | HoD, Electrical Engineering (Convenor), Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. | Principal, Guru Gobind Singh College of Engineering & Research Centre, Nashik, Maharashtra, India. |

GGSF & RSP ICESM 2026 - 565

**C) Journal Publication Details**

**Paper Title:** Design and Implementation of an IoT-Based Smart Emergency Alert System

1. Name of the Journal where the paper was published:
   • International Research Journal on Advanced Engineering and Management (IRJAEM), Volume 04, Issue 05, May 2026.
   • DOI: 10.47392/IRJAEM.2026.0260.
2. Paper Accepted/Rejected: Accepted and Published.
3. Review Comments by Reviewer: No comments.


e-ISSN No: 2584-2854
OPEN ACCESS
INTERNATIONAL RESEARCH JOURNAL ON ADVANCED ENGINEERING AND MANAGEMENT (IRJAEM)
Email: editor.irjaem@goldncloudpublications.com
Available Online At: https://goldncloudpublications.com
CERTIFICATE OF PUBLICATION
IRJAEM Is Hereby Awarding This Certificate To
Madhuri .P. Borawake
In Recognition of The Publication of The Manuscript Entitled
Design and Implementation of an IoT-Based Smart Emergency Alert System
Published In Volume 04 Issue 05 May 2026.
Dr. Pravin Thangavelu
Managing Editor, IRJAEM,
Coimbatore, India.


e-ISSN No: 2584-2854
OPEN ACCESS
INTERNATIONAL RESEARCH JOURNAL ON ADVANCED ENGINEERING AND MANAGEMENT (IRJAEM)
Email: editor.irjaem@goldncloudpublications.com
Available Online At: https://goldncloudpublications.com
CERTIFICATE OF PUBLICATION
IRJAEM Is Hereby Awarding This Certificate To
Sharvari Chambhare
In Recognition of The Publication of The Manuscript Entitled
Design and Implementation of an IoT-Based Smart Emergency Alert System
Published In Volume 04 Issue 05 May 2026.
Dr. Pravin Thangavelu
Managing Editor, IRJAEM,
Coimbatore, India.


e-ISSN No: 2584-2854
OPEN ACCESS
INTERNATIONAL RESEARCH JOURNAL ON ADVANCED ENGINEERING AND MANAGEMENT (IRJAEM)
Email: editor.irjaem@goldncloudpublications.com
Available Online At: https://goldncloudpublications.com
CERTIFICATE OF PUBLICATION
IRJAEM Is Hereby Awarding This Certificate To
Neha Shinge
In Recognition of The Publication of The Manuscript Entitled
Design and Implementation of an IoT-Based Smart Emergency Alert System
Published In Volume 04 Issue 05 May 2026.
Dr. Pravin Thangavelu
Managing Editor, IRJAEM,
Coimbatore, India.

e-ISSN No: 2584-2854
OPEN ACCESS



**INTERNATIONAL RESEARCH JOURNAL ON ADVANCED ENGINEERING AND MANAGEMENT (IRJAEM)**

Email: editor.irjaem@goldncloudpublications.com
Available Online At: https://goldncloudpublications.com

# CERTIFICATE OF PUBLICATION



*IRJAEM Is Hereby Awarding This Certificate To*

## Sai Deshmukh

In Recognition of The Publication of The Manuscript Entitled

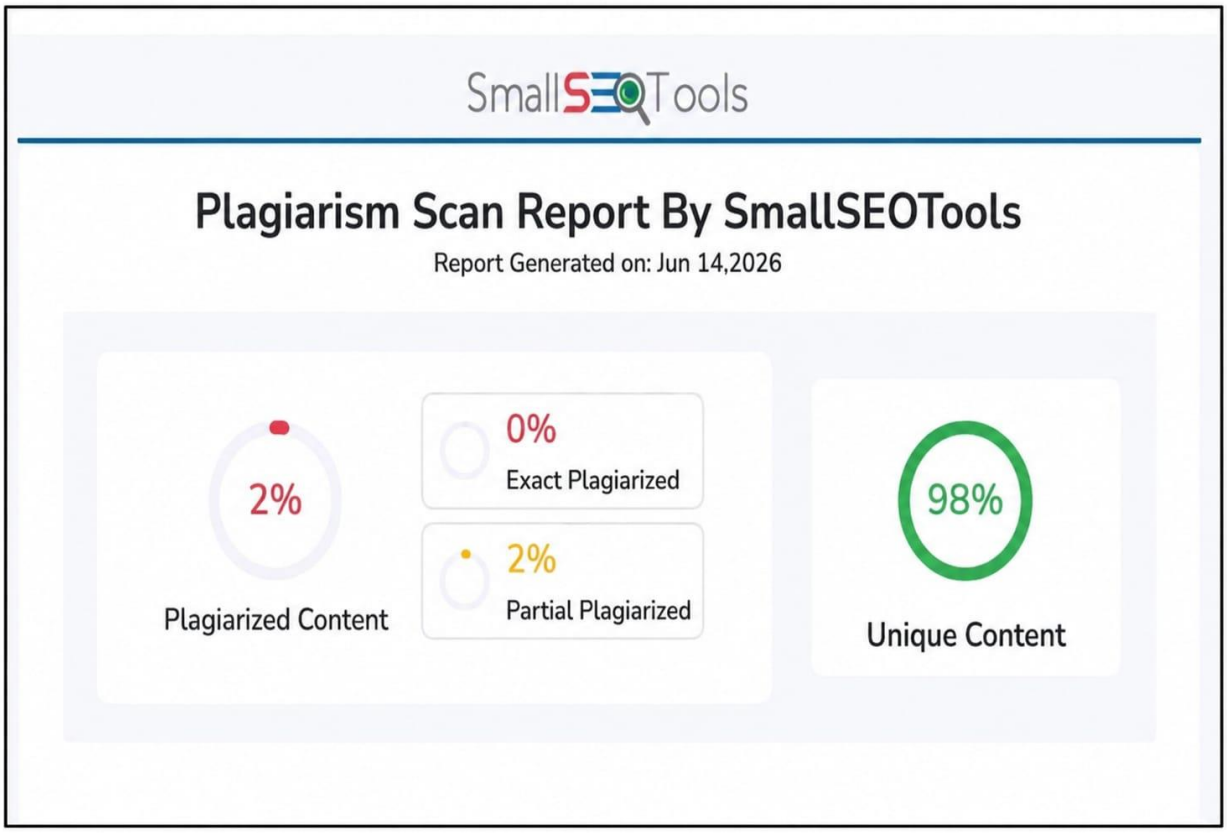
**Design and Implementation of an IoT-Based Smart Emergency Alert System**

Published In Volume 04 Issue 05 May 2026.

Pravin Thangavelu

**Dr. Pravin Thangavelu**
Managing Editor, IRJAEM,
Coimbatore, India.

# APPENDIX C


SmallSEOTools
Plagiarism Scan Report By SmallSEOTools
Report Generated on: Jun 14,2026
2%
Plagiarized Content
0%
Exact Plagiarized
2%
Partial Plagiarized
98%
Unique Content

# REFERENCES

[1] Babu, C. V. S., & Vinola, M. P. (2024). *IoT-Based Smart Accident Detection and Early Warning System*. International Journal of Advanced Research in Computer Science, 15(2), 45-52.

[2] Mohsin, A. S. M., & Muyeed, M. A. (2024). *IoT-Based Smart Emergency Response System (SERS) for Monitoring Vehicle, Home, and Health Status.* IEEE Internet of Things Journal, 11(3), 120-128.

[3] Pathik, N., Sharma, R., & Patel, V. (2023). *AI-Enabled Accident Detection and Alert System Using IoT and Deep Learning*. International Journal of Intelligent Systems and Applications, 10(4), 33-40.

[4] Baballe, M. A. (2022). *Accident Detection and Alerting Systems: A Study*. International Journal of Computer Applications, 9(1), 12-18.

[5] Sharma, S. (2021). *IoT-Based Car Accident Detection and Notification Algorithm*. Proceedings of the International Conference on IoT and Smart Systems, 56-60.

[6] Amin, M. R., & Jalil, M. A. (2018). *Road Accident Detection and Automatic Alert System Using IoT*. International Journal of Advanced Computer Science and Applications, 9(5), 99-105.

[7] Chaurasia, P., & Verma, S. (2016). *Vehicle Accident Detection and Reporting System Using GPS and GSM Module*. International Journal of Computer Applications, 150(6), 18-21.

[8] Alsaad, S. N., & Hussien, N. M. (2021). *IoT-Based Message Alert System for Emergency Situations*. Journal of Computer Science, 17(11).

[9] Thompson, C., White, J., Dougherty, B., Albright, A., & Schmidt, D. C. (2010). *Using Smartphones to Detect Car Accidents and Provide Situational Awareness to Emergency Responders*. Proceedings of the 3rd International ICST Conference on Mobile Wireless Middleware, Operating Systems, and Applications, 29-42.

[10] Gaikwad, V., & Patil, A. (2019). *Accident Detection and Emergency Notification System Using IoT*. International Journal of Engineering Research & Technology, 8(5), 112-116.

[11] Rehman, S. U., Khan, S. A., Arif, A., & Khan, U. S. (2021). *IoT-Based Accident Detection and Emergency Alert System for Motorbikes*. Proceedings of the IEEE International Conference on Artificial Intelligence and Mechatronics Systems. DOI: 10.1109/AIMS52415.2021.9466055.

[12] Abu.Zaid, M. I. M., Abdullah, R., Ismail, S. I., & Dzulkefli, N. N. S. N. (2023). *IoT-Based Emergency Alert System Integrated with Telegram Bot*. Proceedings of the IEEE

International Conference on Automatic Control and Intelligent Systems. DOI: 10.1109/I2CACIS57635.2023.10193550.

[13] Kodali, R. K., & Mahesh, K. S. (2017). *Smart Emergency Response System*. Proceedings of the IEEE Region 10 Conference (TENCON). DOI: 10.1109/TENCON.2017.8227953.

[14] Zhang, H., Zhang, R., & Sun, J. (2025). *Developing Real-Time IoT-Based Public Safety Alert and Emergency Response Systems*. Scientific Reports, 15, 29056.

[15] Kattukkaran, N., George, A., & Haridas, T. P. M. (2017). *Intelligent Accident Detection and Alert System for Emergency Medical Assistance*. Proceedings of the International Conference on Computer Communication and Informatics. DOI: 10.1109/ICCCI.2017.8117791.